26016 Concourse viewing platform
Model v20 - ULS member force envelopes (SOFiSTiK WinGRAF)

Prepared 13 Nov 2026 from 26016_check_v20 (v19m + qp 0.5 kPa for all wind + My hinges at 12 external stair stringer ends), read from a copy of the database.

Envelope: the model's own MAXIMA superposition COMB 3 (EXTR NONL, TYPE DESI, BASE 4300): each of the 43 ULS STR combinations 101-130, 135, 141-148, 151-154 enters as an exclusive alternative (TYPE AG1, factor 1.00), so every envelope value is the value of one complete combination. EQU combinations 131-134 are not included. The combinations are linear (ASE SYST PROB LINE) COPY combinations.

Beam envelope load cases: N 4321/4322, Vy 4323/4324, Vz 4325/4326, Mt 4327/4328, My 4329/4330, Mz 4331/4332 (max/min). SLS characteristic 201-221: 4421-4432 (same order).

Check: at every result station of every beam the MAXIMA value equals the largest / smallest value read from the 43 combinations (largest difference over all stations: N 0.0e+00, My 1.8e-09 kNm).

Sign convention: SOFiSTiK local beam axes, N > 0 tension, My about the local y (strong) axis, Mz about the local z (weak) axis. Red = max load case, blue = min load case. Units kN, kNm.

Floor framing is drawn in a steep isometric view because My, Vz act in the vertical plane and cannot be read in a true top view (tested: diagrams rotated into the plan flip side with the member direction). Mz of Level 1 is drawn in a true plan view (X right, Y up).

Kingpost / central mast: modelled with shell elements (QUAD), no beam group, so no beam N / My diagram is given here.

ULS envelope per member group - forces N, Vy, Vz

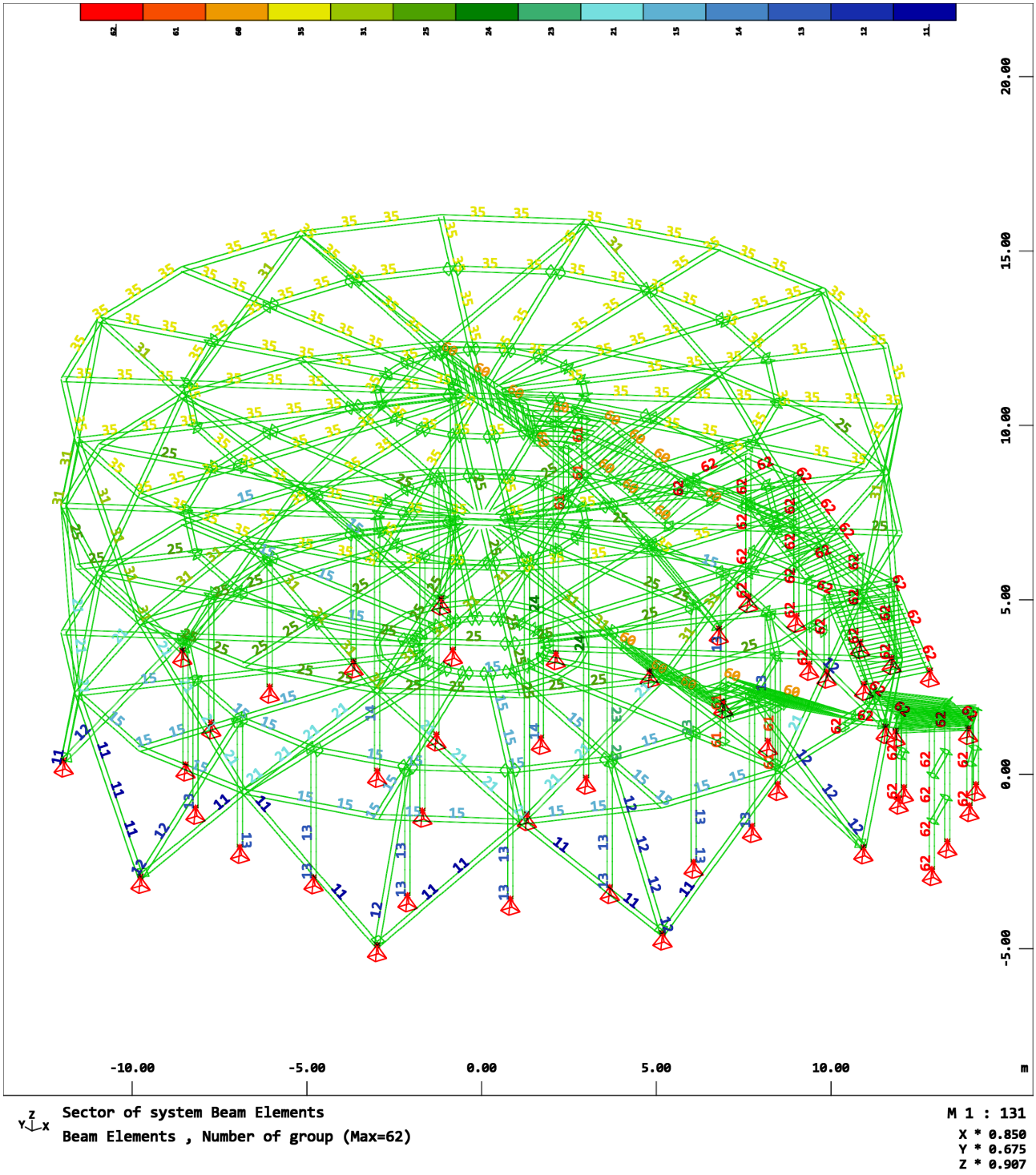
Each cell: envelope value, governing combination (LC), beam element and x [m from beam start]. Full data incl. SLN: env_table_ULS.csv / .json.

Group							
11	V-column rakers, ground to L1 SHS 150x150x10	18.5 (LC 109) b 60196 x 0.42	-290.6 (LC 147) b 60239 x 0.00	1.4 (LC 120) b 60141 x 0.00	-1.6 (LC 126) b 60084 x 0.42	2.6 (LC 120) b 60071 x 0.00	-2.7 (LC 122) b 60154 x 0.42
12	rakers, ground to L1 SHS 150x150x10	37.7 (LC 116) b 70104 x 0.41	-76.5 (LC 105) b 70053 x 0.00	0.8 (LC 130) b 70079 x 0.00	-1.1 (LC 125) b 70092 x 0.00	1.4 (LC 115) b 70001 x 0.00	-1.4 (LC 123) b 70013 x 0.41
13	vertical columns (ground-L1, L1-L2) SHS 150x150x10	10.6 (LC 110) b 80061 x 0.00	-264.8 (LC 121) b 80050 x 0.41	6.1 (LC 124) b 80011 x 0.00	-4.8 (LC 124) b 80030 x 0.41	5.1 (LC 123) b 80141 x 0.00	-5.0 (LC 128) b 80151 x 0.00
14	vertical columns, ground to L1 SHS 150x150x10	-20.7 (LC 109) b 90011 x 0.00	-398.8 (LC 146) b 90030 x 0.41	0.8 (LC 152) b 90051 x 0.00	-1.4 (LC 151) b 90011 x 0.00	0.3 (LC 113) b 90010 x 0.41	-0.2 (LC 112) b 90060 x 0.41
15	Level 1 framing (radials + rings) UB 254x146x43	109.2 (LC 135) b 100507 x 0.00	-57.9 (LC 104) b 100333 x 0.00	55.4 (LC 122) b 100841 x 0.25	-57.4 (LC 152) b 100859 x 0.00	76.3 (LC 153) b 100842 x 0.00	-76.0 (LC 153) b 100868 x 0.39
21	V-column rakers, L1 to L2 SHS 150x150x10	29.6 (LC 109) b 110085 x 0.00	-229.6 (LC 145) b 110238 x 0.41	4.5 (LC 120) b 110239 x 0.00	-4.4 (LC 120) b 110014 x 0.41	7.2 (LC 121) b 110113 x 0.00	-6.4 (LC 119) b 110196 x 0.41
23	vertical columns, L1 to L2 SHS 150x150x10	-6.3 (LC 109) b 120001 x 0.00	-146.1 (LC 121) b 120020 x 0.40	3.6 (LC 154) b 120001 x 0.00	-4.1 (LC 152) b 120020 x 0.40	0.6 (LC 128) b 120010 x 0.40	-0.4 (LC 123) b 120020 x 0.40
24	vertical columns, L1 to L2 SHS 150x150x10	-5.3 (LC 109) b 130021 x 0.00	-263.8 (LC 146) b 130040 x 0.40	4.6 (LC 110) b 130021 x 0.00	-8.4 (LC 125) b 130060 x 0.40	5.2 (LC 110) b 130040 x 0.40	-5.5 (LC 124) b 130060 x 0.40
25	Level 2 framing (radials + rings) UB 254x146x43; UB 457x191x89	62.5 (LC 145) b 140632 x 0.00	-33.2 (LC 110) b 140748 x 0.00	32.0 (LC 152) b 140775 x 0.00	-29.4 (LC 119) b 140439 x 0.00	146.0 (LC 146) b 140684 x 0.00	-109.4 (LC 145) b 140389 x 0.39
31	V-column rakers, L2 to roof SHS 150x150x10	45.3 (LC 112) b 150085 x 0.00	-59.9 (LC 147) b 150099 x 0.00	5.8 (LC 145) b 150196 x 0.41	-5.9 (LC 145) b 150169 x 0.00	7.2 (LC 153) b 150127 x 0.00	-6.9 (LC 145) b 150042 x 0.41
35	roof framing (radials + rings) UB 254x146x43	54.5 (LC 113) b 160070 x 0.42	-53.4 (LC 112) b 160030 x 0.42	1.2 (LC 125) b 160129 x 0.42	-1.1 (LC 126) b 160070 x 0.42	19.1 (LC 110) b 160627 x 0.39	-19.3 (LC 108) b 160081 x 0.00
60	internal stair (stringers, landing, treads) RHS 200x100x6; SHS 50x50x5; H = 5 mm	67.5 (LC 152) b 170329 x 0.34	-24.1 (LC 135) b 170316 x 0.00	24.0 (LC 122) b 170330 x 0.30	-31.4 (LC 152) b 170138 x 0.00	38.2 (LC 121) b 170306 x 0.00	-32.5 (LC 152) b 170008 x 0.00
61	internal stair landing posts SHS 100x100x5	0.3 (LC 107) b 180001 x 0.00	-50.9 (LC 120) b 180010 x 0.40	0.0 (LC 135) b 180014 x 0.00	-0.0 (LC 135) b 180003 x 0.00	0.0 (LC 135) b 180011 x 0.00	-0.0 (LC 127) b 180018 x 0.00
62	external stair ground to L1 RHS 200x100x6; H = 5 mm; SHS 100x100x5; SHS 80x80x6	49.2 (LC 111) b 190321 x 0.05	-54.5 (LC 135) b 190321 x 0.00	27.4 (LC 135) b 190016 x 0.00	-29.0 (LC 135) b 190161 x 0.00	13.8 (LC 121) b 190459 x 0.00	-14.9 (LC 135) b 190016 x 0.00

ULS envelope per member group - moments Mt, My, Mz

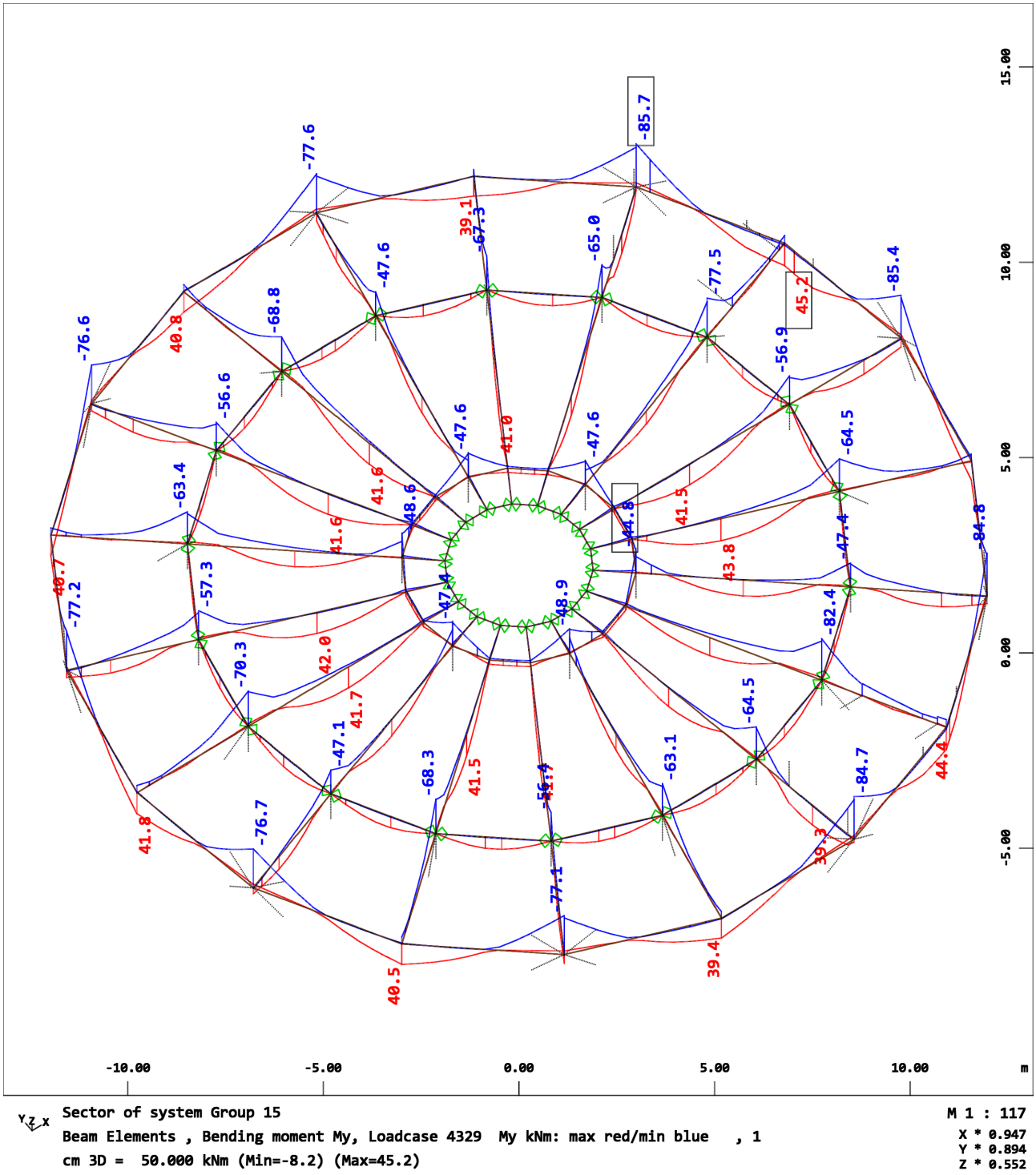
Each cell: envelope value, governing combination (LC), beam element and x [m from beam start]. Full data incl. SLN: env_table_ULS.csv / .json.

Group							
11	V-column rakers, ground to L1 SHS 150x150x10	1.1 (LC 152) b 60167 x 0.00	-1.1 (LC 152) b 60058 x 0.00	4.2 (LC 122) b 60140 x 0.42	-5.8 (LC 151) b 60113 x 0.00	5.8 (LC 152) b 60084 x 0.42	-2.2 (LC 152) b 60071 x 0.00
12	rakers, ground to L1 SHS 150x150x10	0.4 (LC 124) b 70074 x 0.00	-0.4 (LC 125) b 70010 x 0.00	1.8 (LC 115) b 70007 x 0.41	-0.0 (LC 118) b 70078 x 0.41	2.3 (LC 125) b 70104 x 0.41	-1.7 (LC 126) b 70078 x 0.41
13	vertical columns (ground-L1, L1-L2) SHS 150x150x10	0.2 (LC 112) b 80183 x 0.00	-0.7 (LC 152) b 80200 x 0.00	4.6 (LC 118) b 80015 x 0.41	-4.6 (LC 125) b 80015 x 0.41	9.6 (LC 152) b 80200 x 0.40	-12.2 (LC 152) b 80021 x 0.00
14	vertical columns, ground to L1 SHS 150x150x10	0.0 (LC 151) b 90005 x 0.00	-0.0 (LC 151) b 90023 x 0.00	0.7 (LC 112) b 90051 x 0.00	-1.1 (LC 113) b 90001 x 0.00	3.5 (LC 152) b 90051 x 0.00	-5.5 (LC 151) b 90011 x 0.00
15	Level 1 framing (radials + rings) UB 254x146x43	1.3 (LC 135) b 100859 x 0.00	-1.2 (LC 135) b 100841 x 0.00	45.2 (LC 152) b 100790 x 0.00	-85.7 (LC 152) b 100832 x 0.39	11.2 (LC 120) b 100841 x 0.00	-11.6 (LC 120) b 100859 x 0.00
21	V-column rakers, L1 to L2 SHS 150x150x10	2.2 (LC 122) b 110011 x 0.00	-2.2 (LC 121) b 110251 x 0.00	20.9 (LC 135) b 110140 x 0.41	-22.3 (LC 121) b 110113 x 0.00	9.3 (LC 121) b 110155 x 0.00	-15.9 (LC 119) b 110252 x 0.41
23	vertical columns, L1 to L2 SHS 150x150x10	0.7 (LC 152) b 120015 x 0.00	-0.1 (LC 125) b 120001 x 0.00	2.2 (LC 128) b 120010 x 0.40	-1.7 (LC 123) b 120020 x 0.40	10.1 (LC 152) b 120020 x 0.40	-6.5 (LC 151) b 120020 x 0.40
24	vertical columns, L1 to L2 SHS 150x150x10	0.1 (LC 129) b 130011 x 0.00	-0.1 (LC 111) b 130041 x 0.00	3.9 (LC 130) b 130040 x 0.40	-4.1 (LC 123) b 130030 x 0.40	10.6 (LC 153) b 130020 x 0.40	-13.4 (LC 153) b 130051 x 0.00
25	Level 2 framing (radials + rings) UB 254x146x43; UB 457x191x89	1.0 (LC 120) b 140490 x 0.00	-1.2 (LC 148) b 140489 x 0.00	233.7 (LC 120) b 140698 x 0.39	-363.8 (LC 145) b 140389 x 0.39	12.3 (LC 152) b 140769 x 0.41	-10.4 (LC 121) b 140775 x 0.60
31	V-column rakers, L2 to roof SHS 150x150x10	3.9 (LC 111) b 150190 x 0.00	-3.8 (LC 111) b 150172 x 0.00	21.1 (LC 104) b 150070 x 0.41	-23.7 (LC 153) b 150127 x 0.00	19.0 (LC 145) b 150182 x 0.41	-15.0 (LC 145) b 150169 x 0.00
35	roof framing (radials + rings) UB 254x146x43	0.0 (LC 112) b 160815 x 0.00	-0.0 (LC 108) b 160782 x 0.00	44.7 (LC 107) b 160735 x 0.39	-41.1 (LC 146) b 160735 x 0.39	1.7 (LC 122) b 160070 x 0.42	-1.7 (LC 121) b 160119 x 0.42
60	internal stair (stringers, landing, treads) RHS 200x100x6; SHS 50x50x5; H = 5 mm	5.2 (LC 135) b 170136 x 0.00	-4.9 (LC 135) b 170001 x 0.00	16.8 (LC 121) b 170002 x 0.00	-22.4 (LC 120) b 170305 x 0.16	8.3 (LC 120) b 170328 x 0.00	-14.5 (LC 122) b 170137 x 0.00
61	internal stair landing posts SHS 100x100x5	0.3 (LC 121) b 180006 x 0.00	-0.2 (LC 122) b 180016 x 0.00	0.0 (LC 124) b 180014 x 0.40	-0.0 (LC 121) b 180007 x 0.00	0.0 (LC 152) b 180004 x 0.40	-0.0 (LC 152) b 180014 x 0.40
62	external stair ground to L1 RHS 200x100x6; H = 5 mm; SHS 100x100x5; SHS 80x80x6	7.3 (LC 135) b 190173 x 0.00	-7.0 (LC 135) b 190028 x 0.00	9.6 (LC 105) b 190465 x 0.00	-10.4 (LC 119) b 190022 x 0.44	6.7 (LC 135) b 190016 x 0.00	-9.2 (LC 135) b 190161 x 0.00



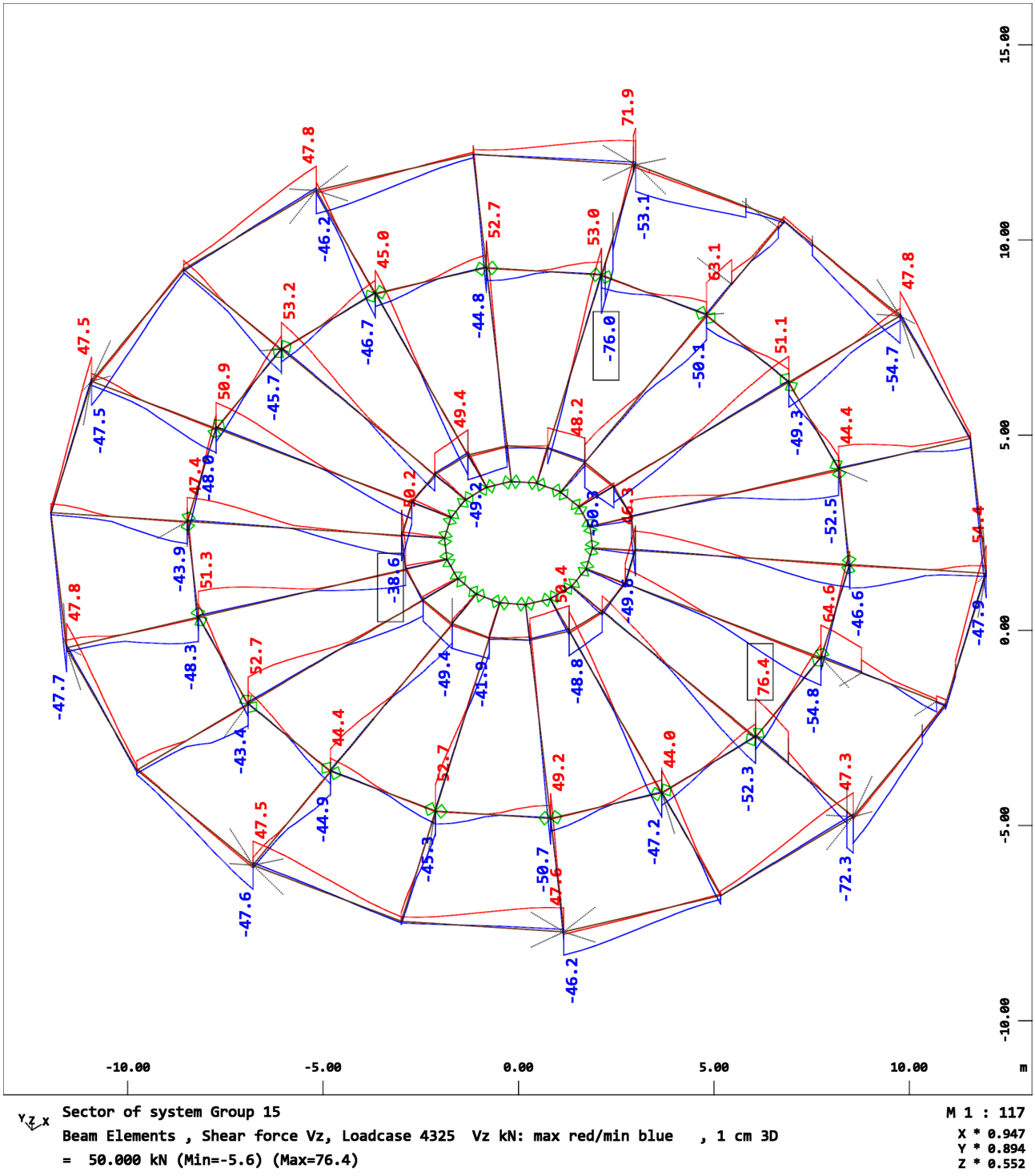
P00_key_beam_groups

Key: beam groups of model v20 (group number on each member, colour legend at top). Groups 1, 2, 6, 7 are shell (QUAD) groups and are not shown.



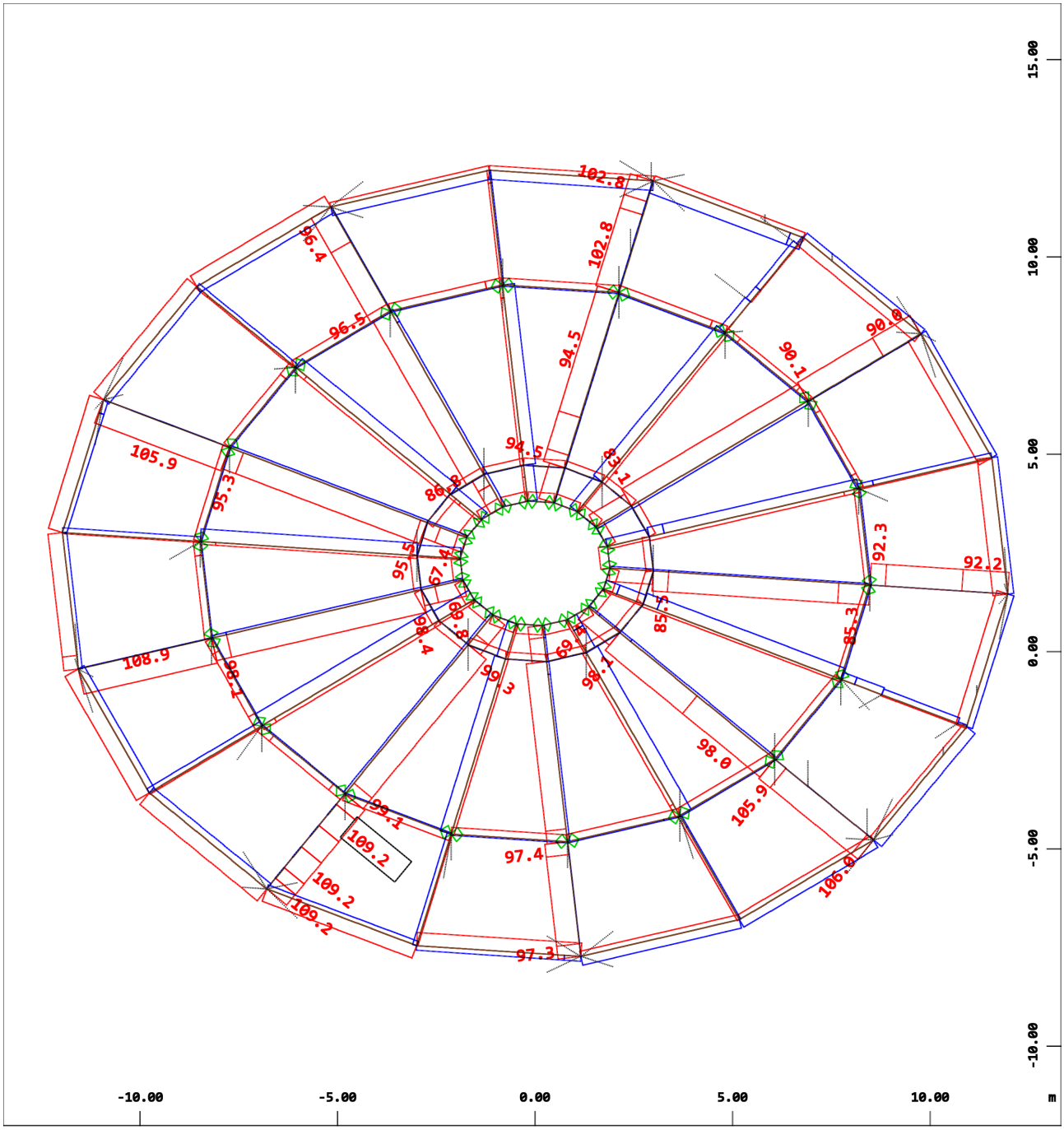
P01_L1_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) My. Diagram scale 1 cm = 50 kNm (on the 25.5 cm sheet); labels only where |value| > 39 kNm. Envelope extremes: max 45.19, min -85.67 kNm.



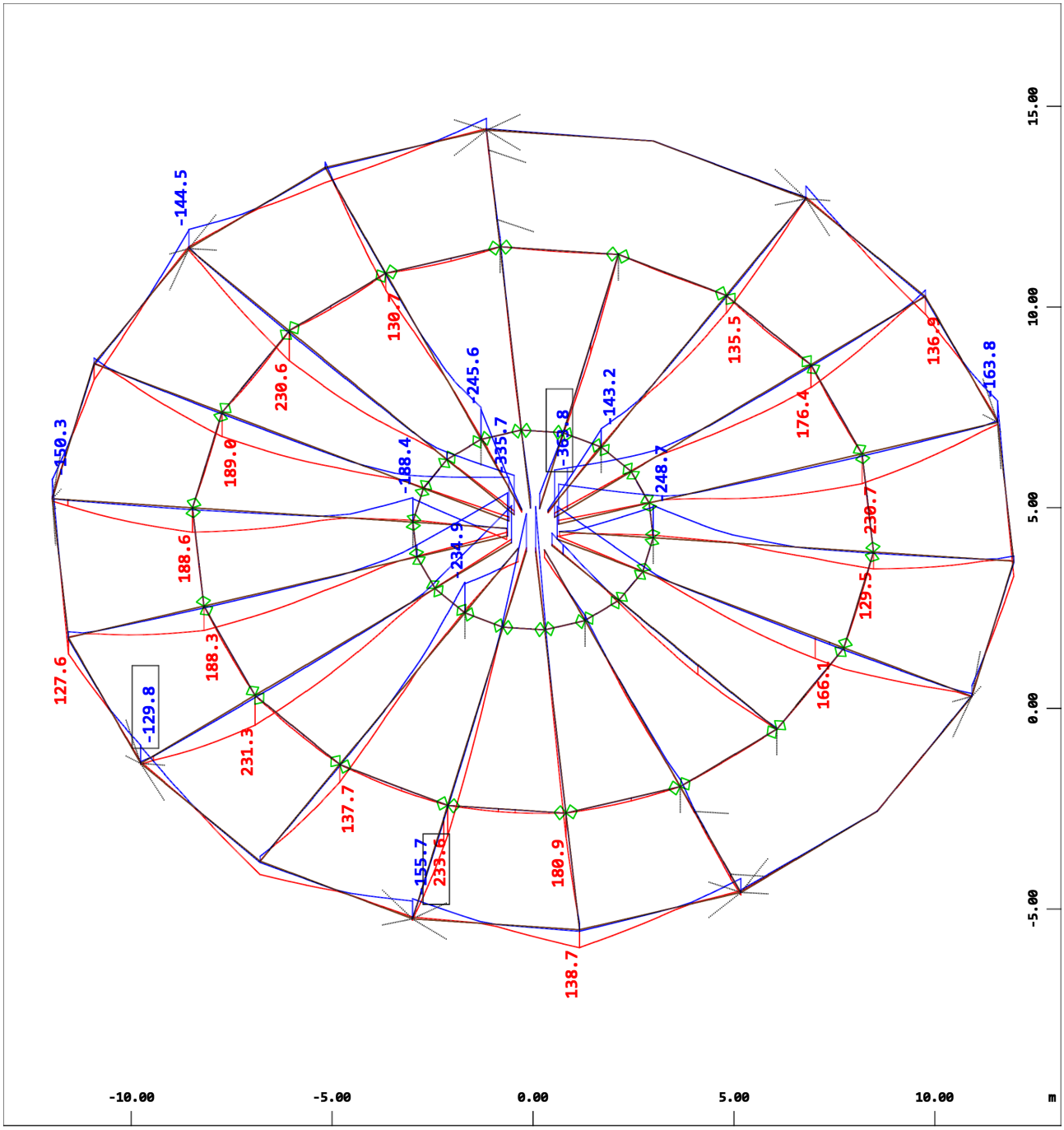
P02_L1_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) Vz. Diagram scale 1 cm = 50 kN (on the 25.5 cm sheet); labels only where |value| > 34 kN. Envelope extremes: max 76.35, min -75.97 kN.



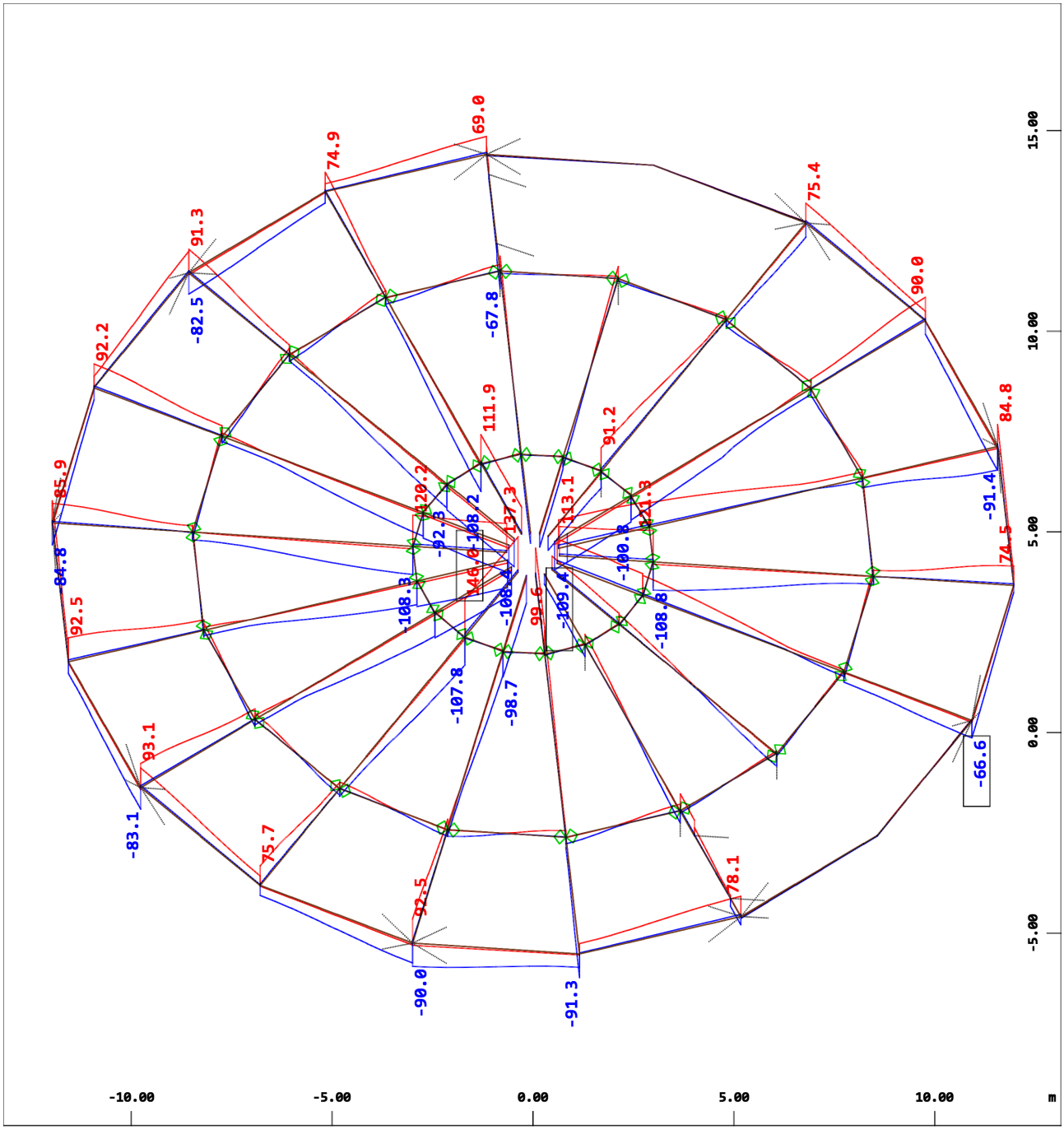
P03_L1_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 66 kN. Envelope extremes: max 109.19, min -57.9 kN.



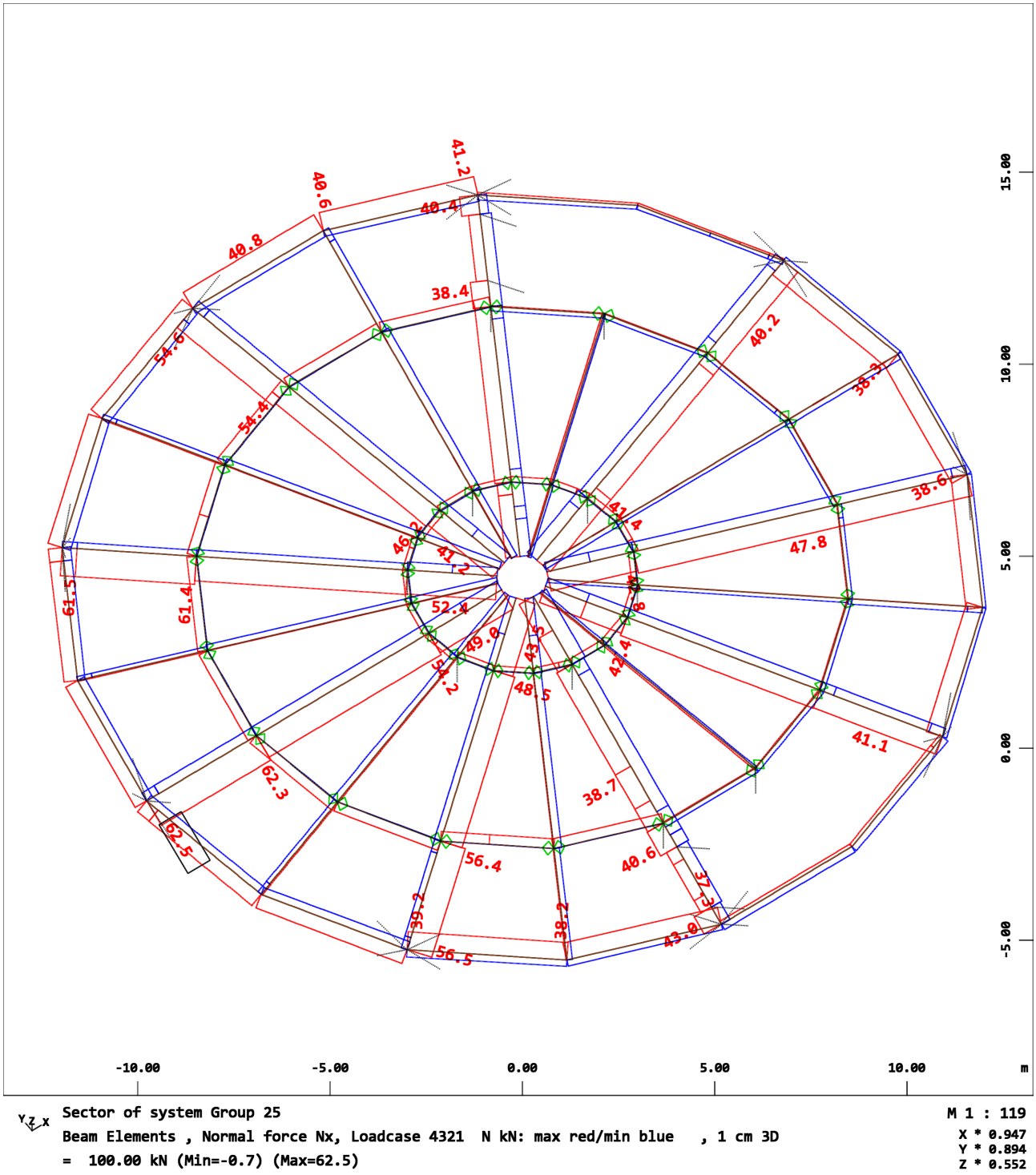
P04_L2_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 2 framing (group 25) My. Diagram scale 1 cm = 200 kNm (on the 25.5 cm sheet); labels only where |value| > 127 kNm. Envelope extremes: max 233.65, min -363.8 kNm.



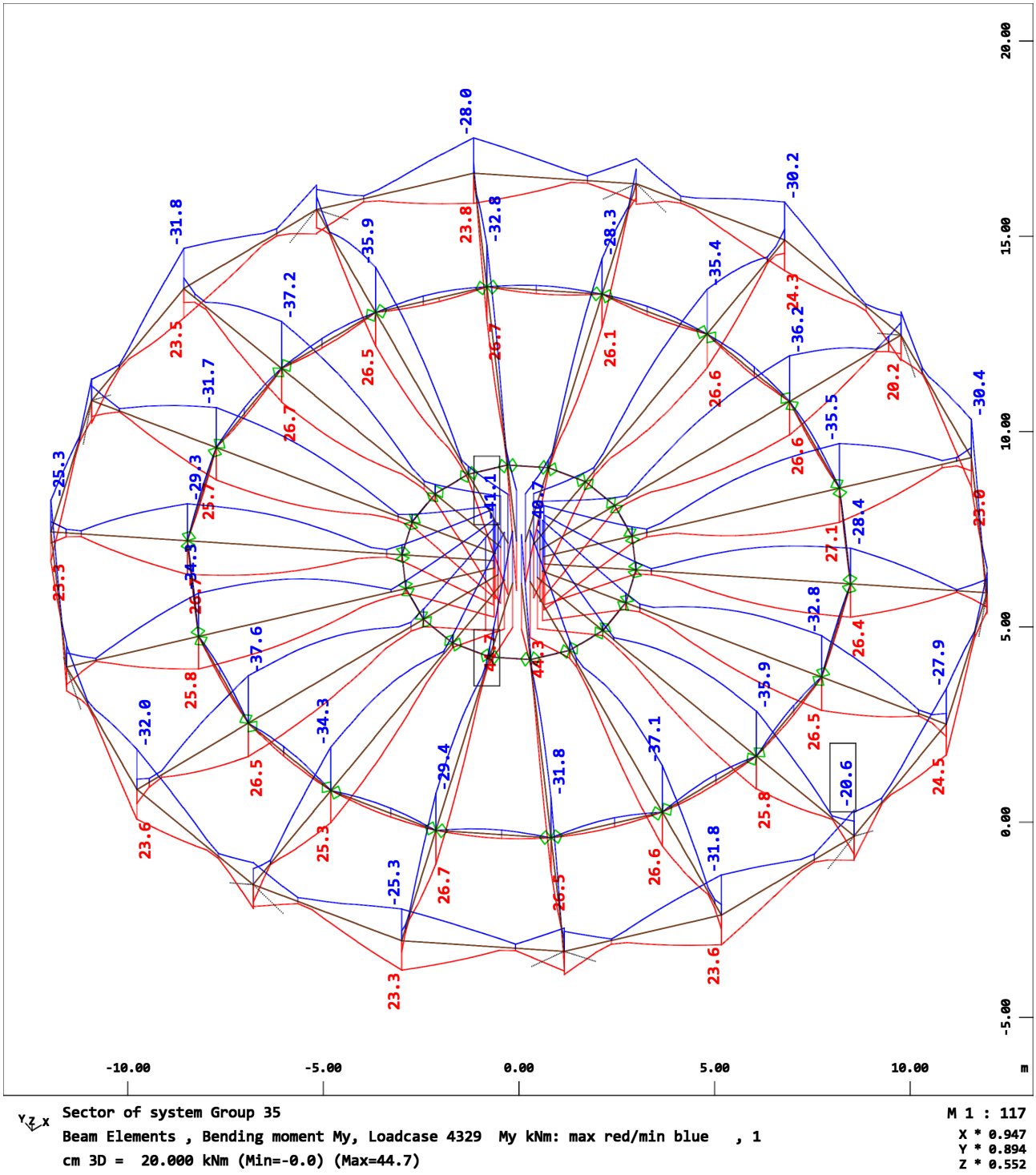
P05_L2_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 2 framing (group 25) Vz. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 66 kN. Envelope extremes: max 145.99, min -109.44 kN.



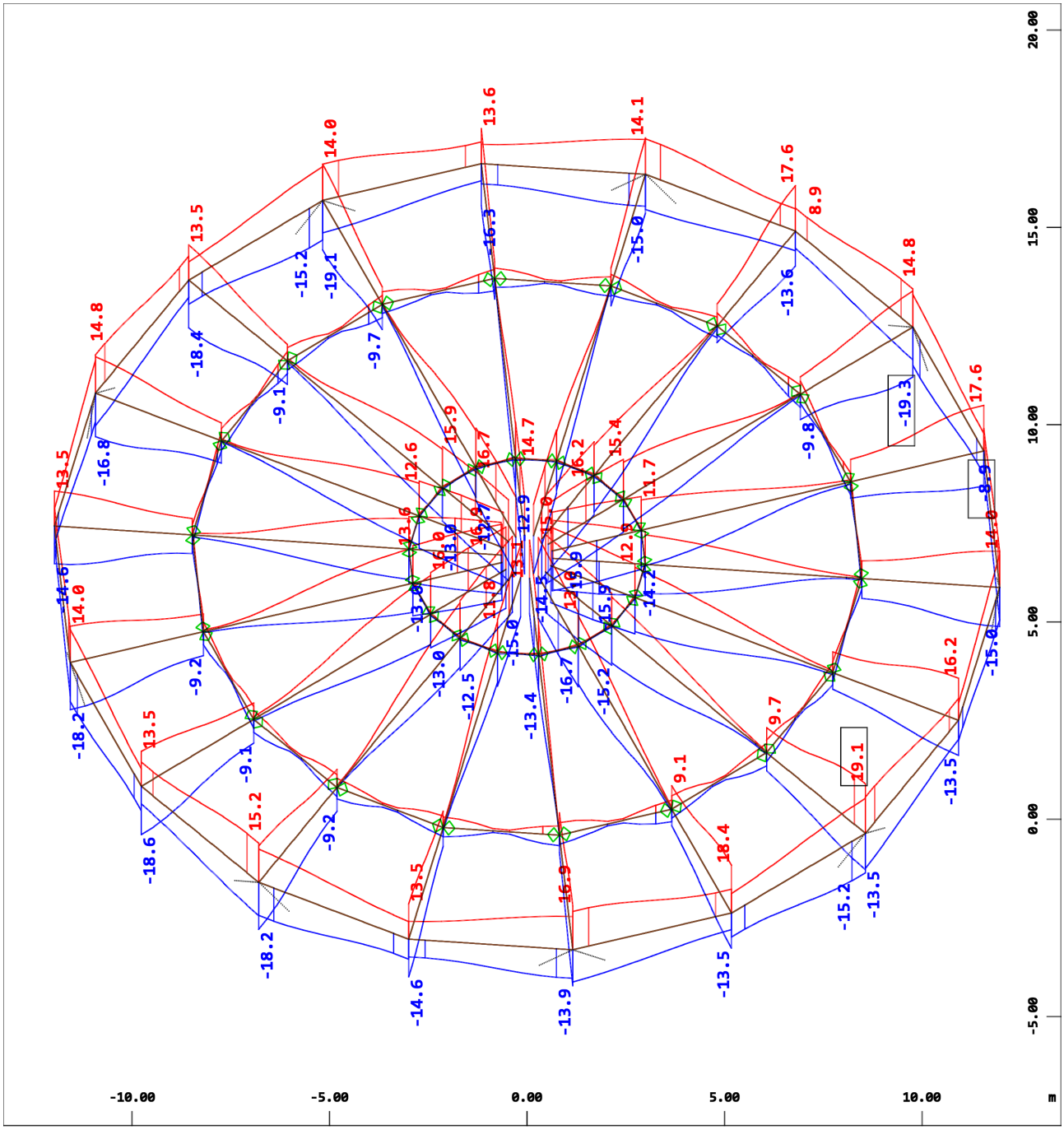
P06_L2_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 2 framing (group 25) N. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 37 kN. Envelope extremes: max 62.45, min -33.18 kN.



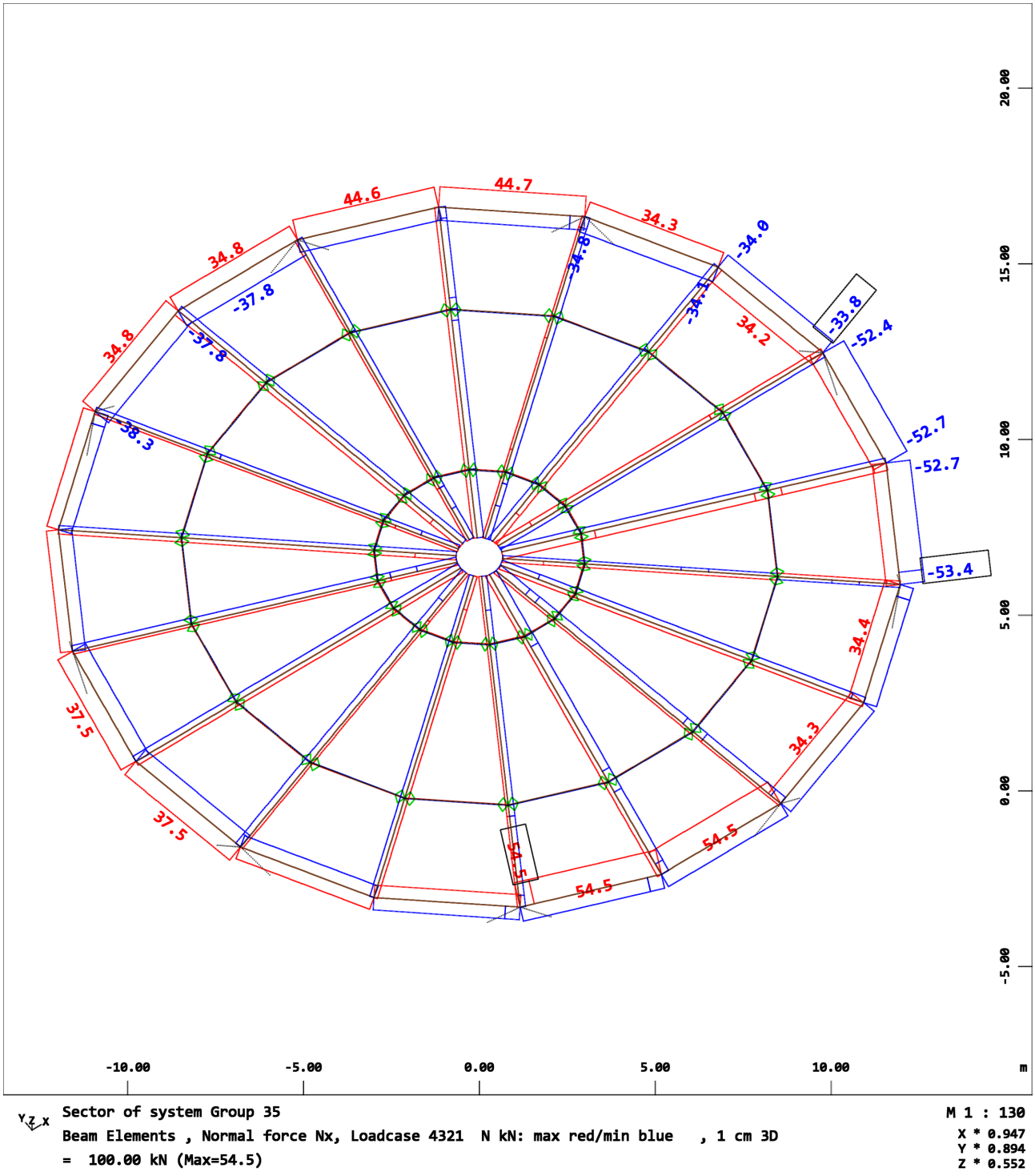
P07_RF_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Roof framing (group 35) My. Diagram scale 1 cm = 20 kNm (on the 25.5 cm sheet); labels only where |value| > 20 kNm. Envelope extremes: max 44.67, min -41.12 kNm.



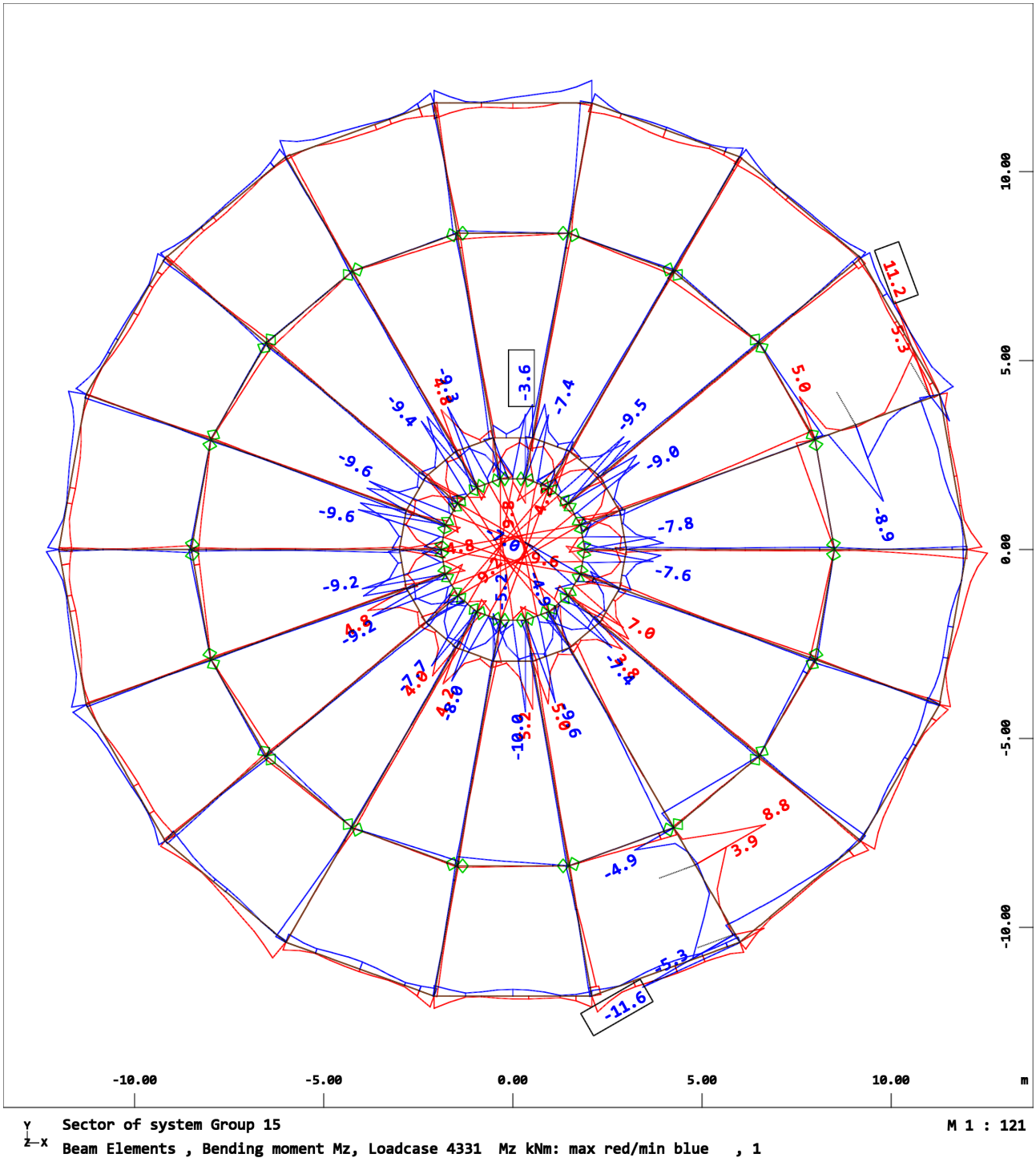
P08_RF_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Roof framing (group 35) Vz. Diagram scale 1 cm = 10 kN (on the 25.5 cm sheet); labels only where |value| > 8.7 kN. Envelope extremes: max 19.08, min -19.28 kN.



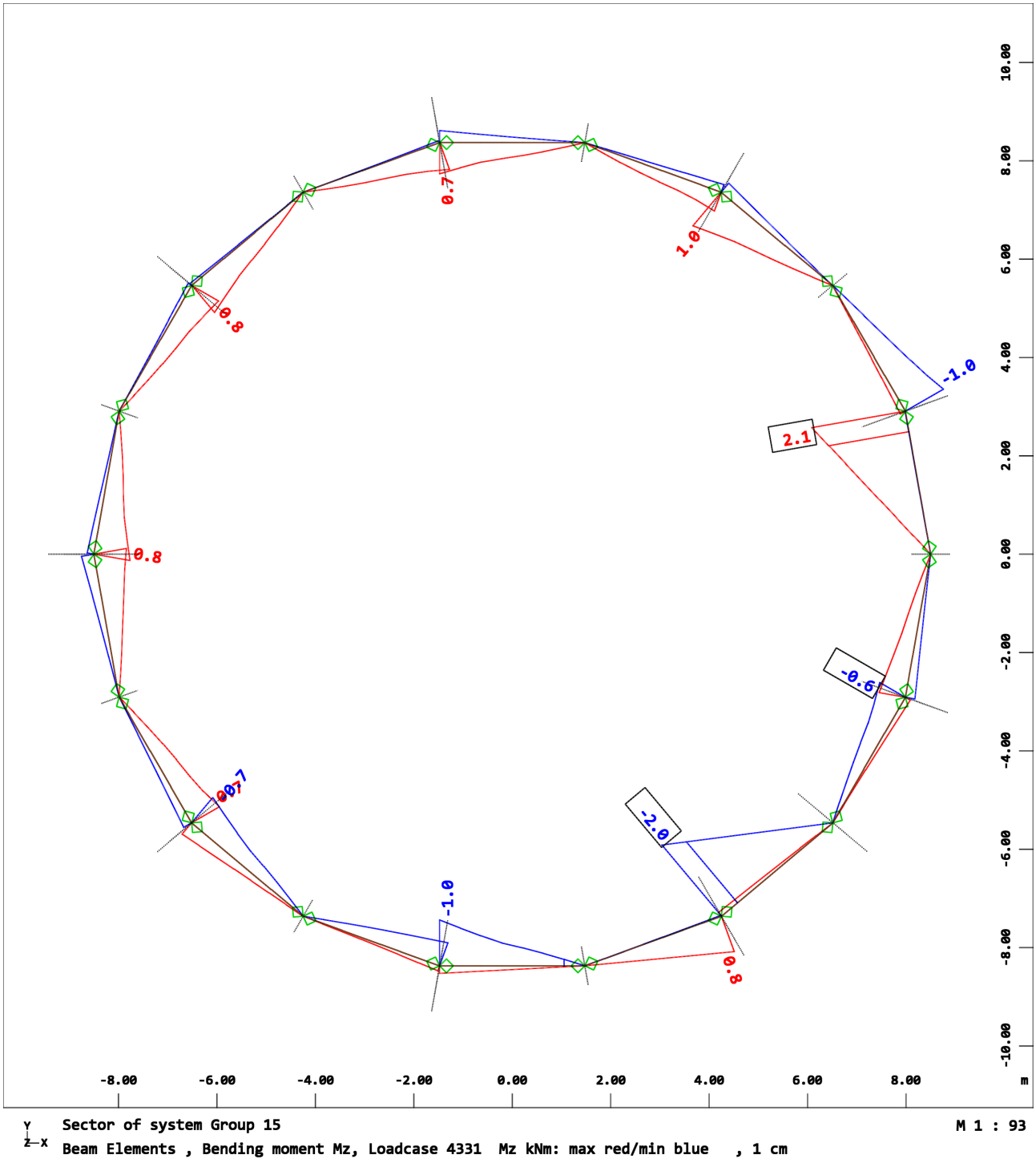
P09_RF_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Roof framing (group 35) N. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 33 kN. Envelope extremes: max 54.53, min -53.4 kN.



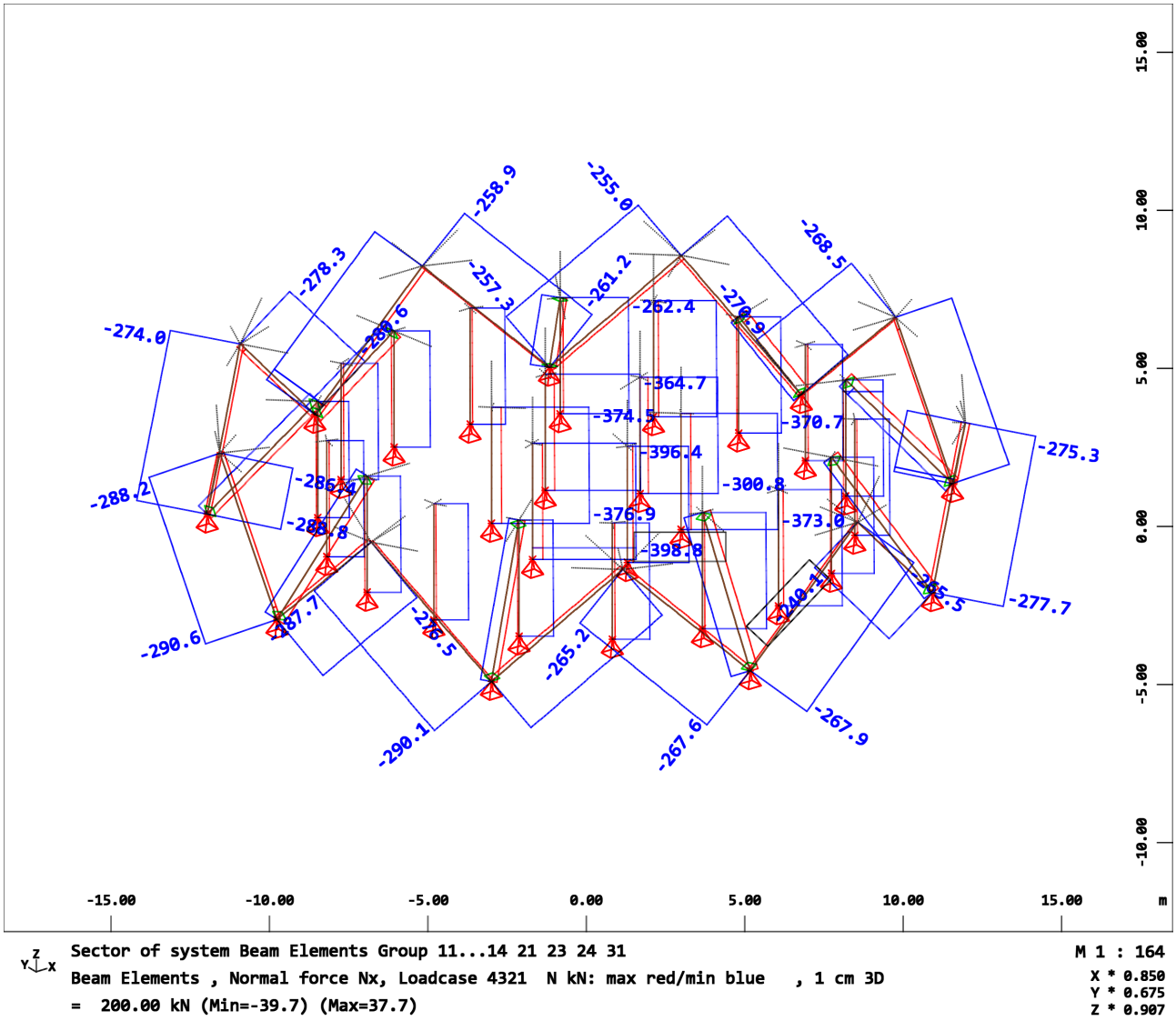
P10_L1_Mz_plan

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) Mz, plan. Diagram scale 1 cm = 5 kNm (on the 25.5 cm sheet); labels only where |value| > 3.5 kNm. Envelope extremes: max 11.22, min -11.64 kNm.



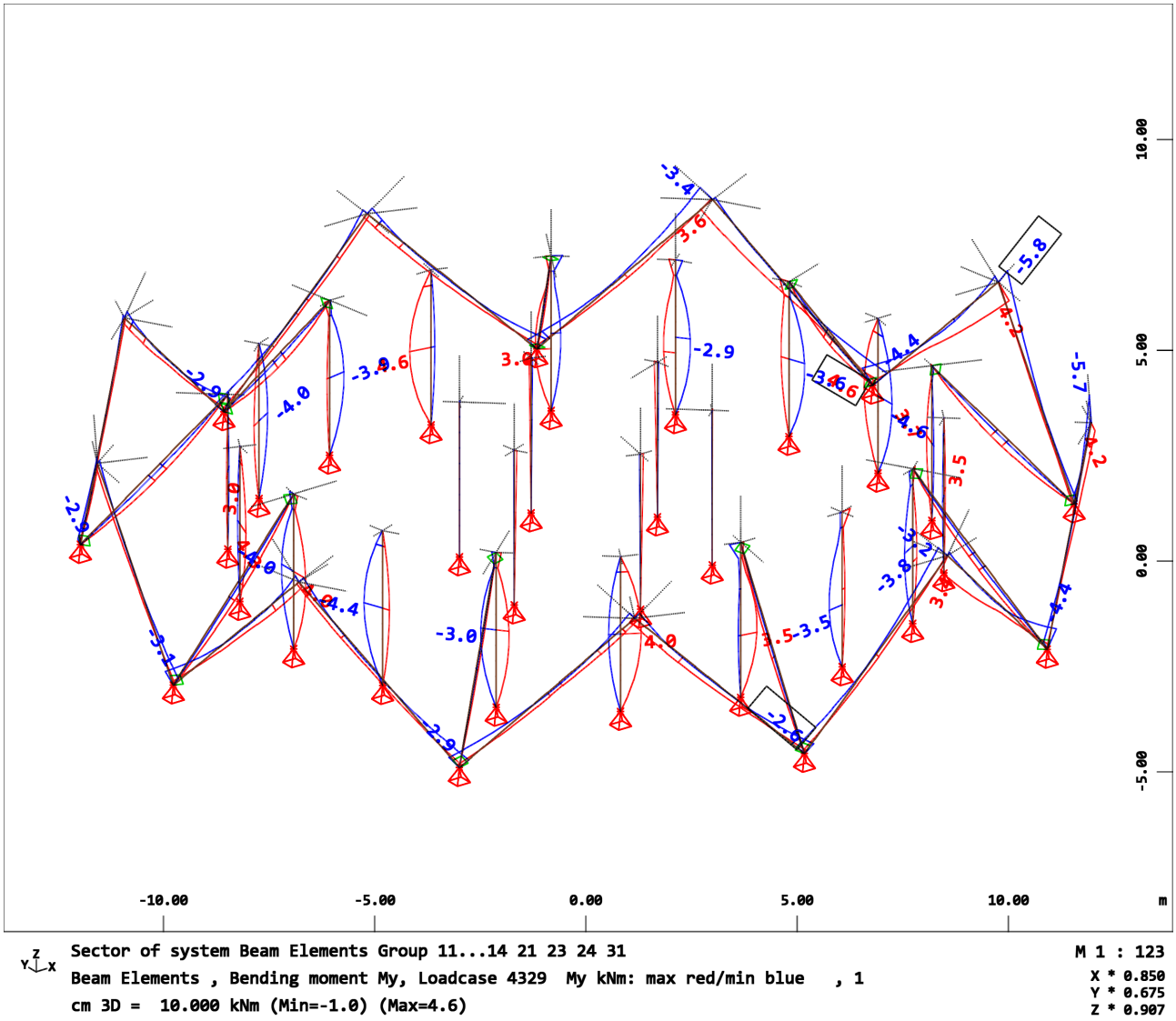
P11_L1_ring85_Mz_plan

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 ring r 8.5 only (group 15) Mz, plan. Diagram scale 1 cm = 1 kNm (on the 25.5 cm sheet); labels only where |value| > 0.6 kNm. Envelope extremes: max 2.07, min -2.03 kNm.



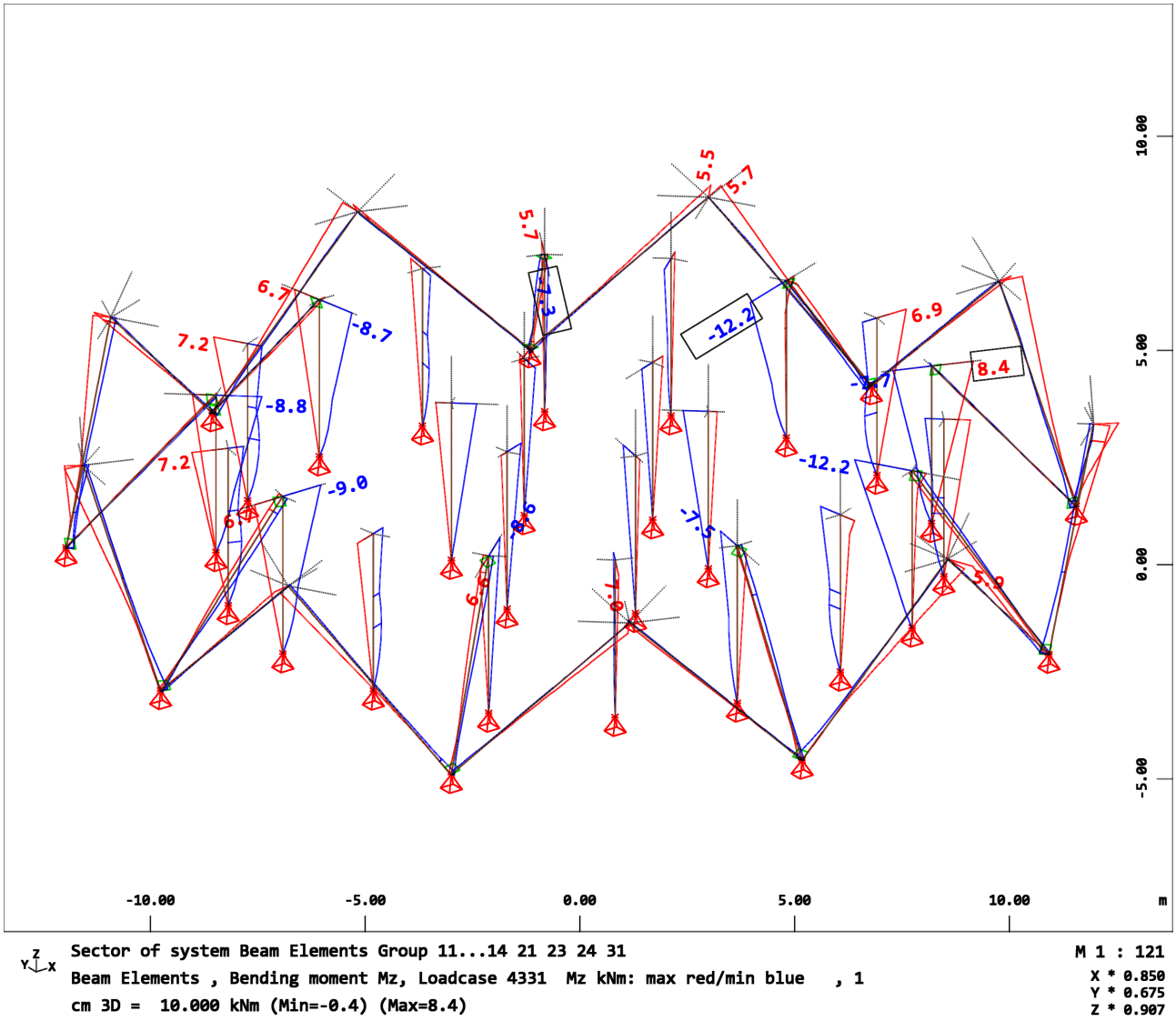
P12_V_G-L1_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers ground to Level 1 (groups 11, 12, 13, 14, 21, 23, 24, 31) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 239 kN. Envelope extremes: max 37.71, min -398.77 kN.



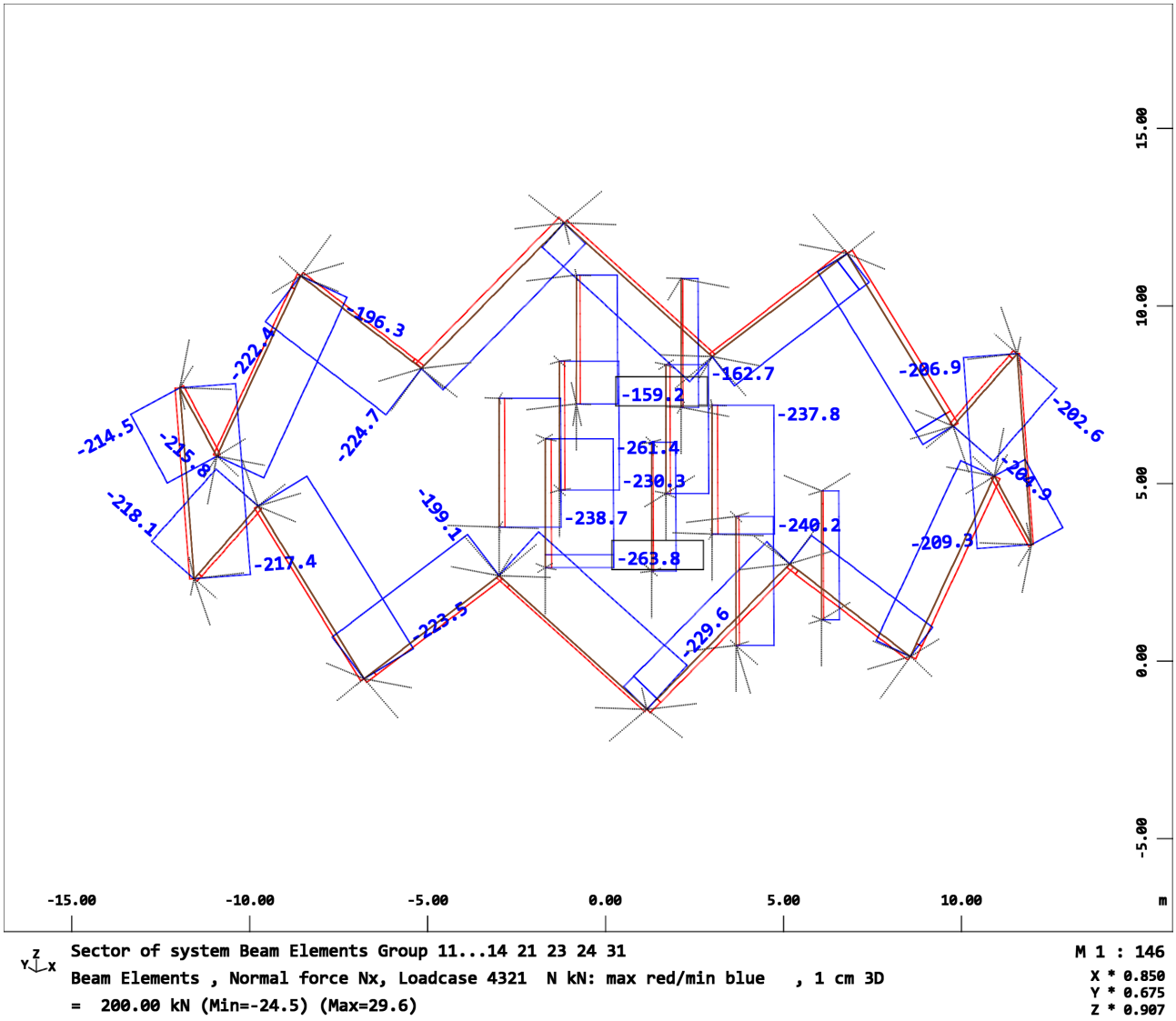
P13_V_G-L1_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers ground to Level 1 (groups 11, 12, 13, 14, 21, 23, 24, 31) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 2.6 kNm. Envelope extremes: max 4.62, min -5.79 kNm.



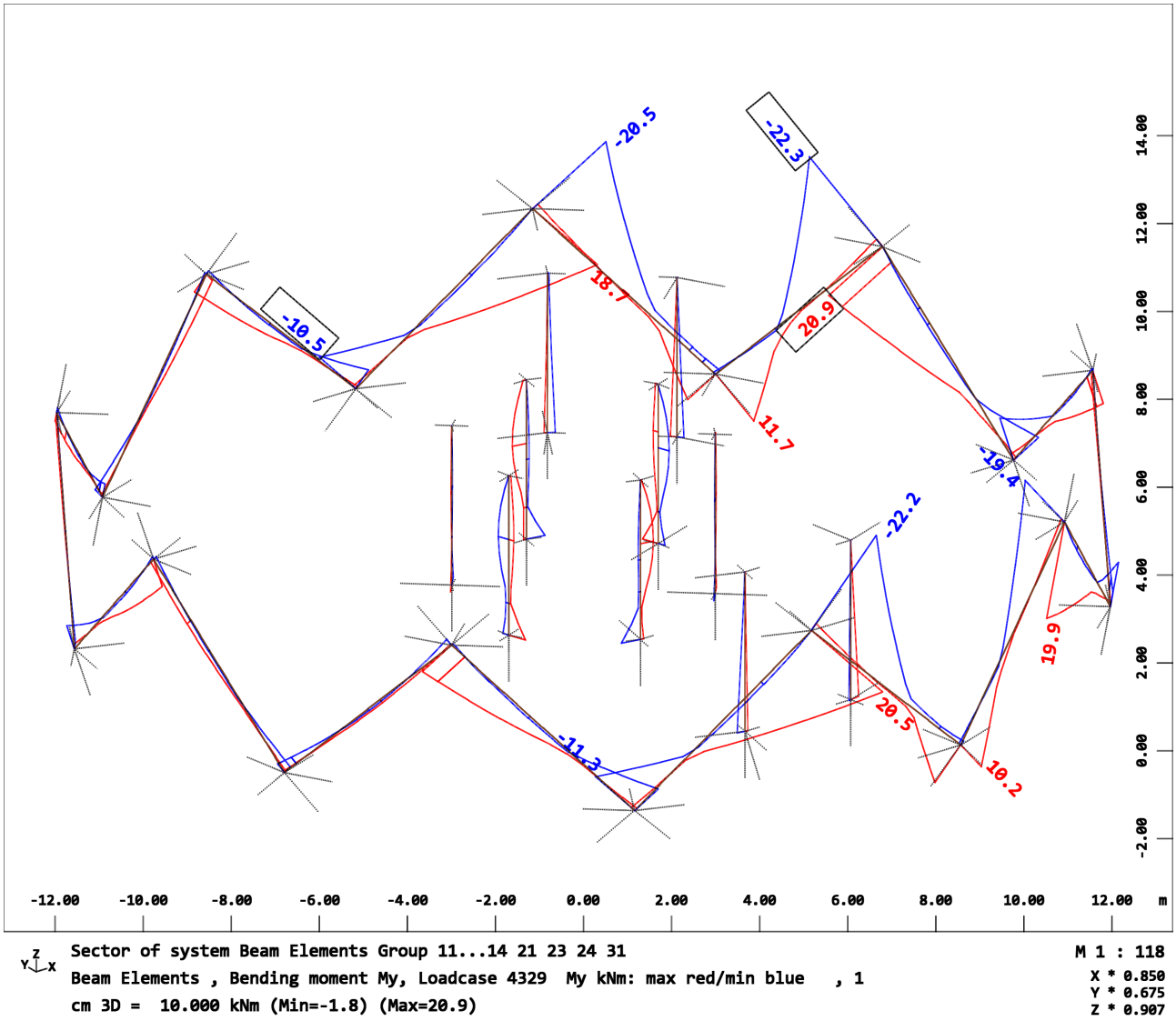
P14_V_G-L1_Mz

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers ground to Level 1 (groups 11, 12, 13, 14, 21, 23, 24, 31) Mz. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 5.5 kNm. Envelope extremes: max 8.35, min -12.24 kNm.



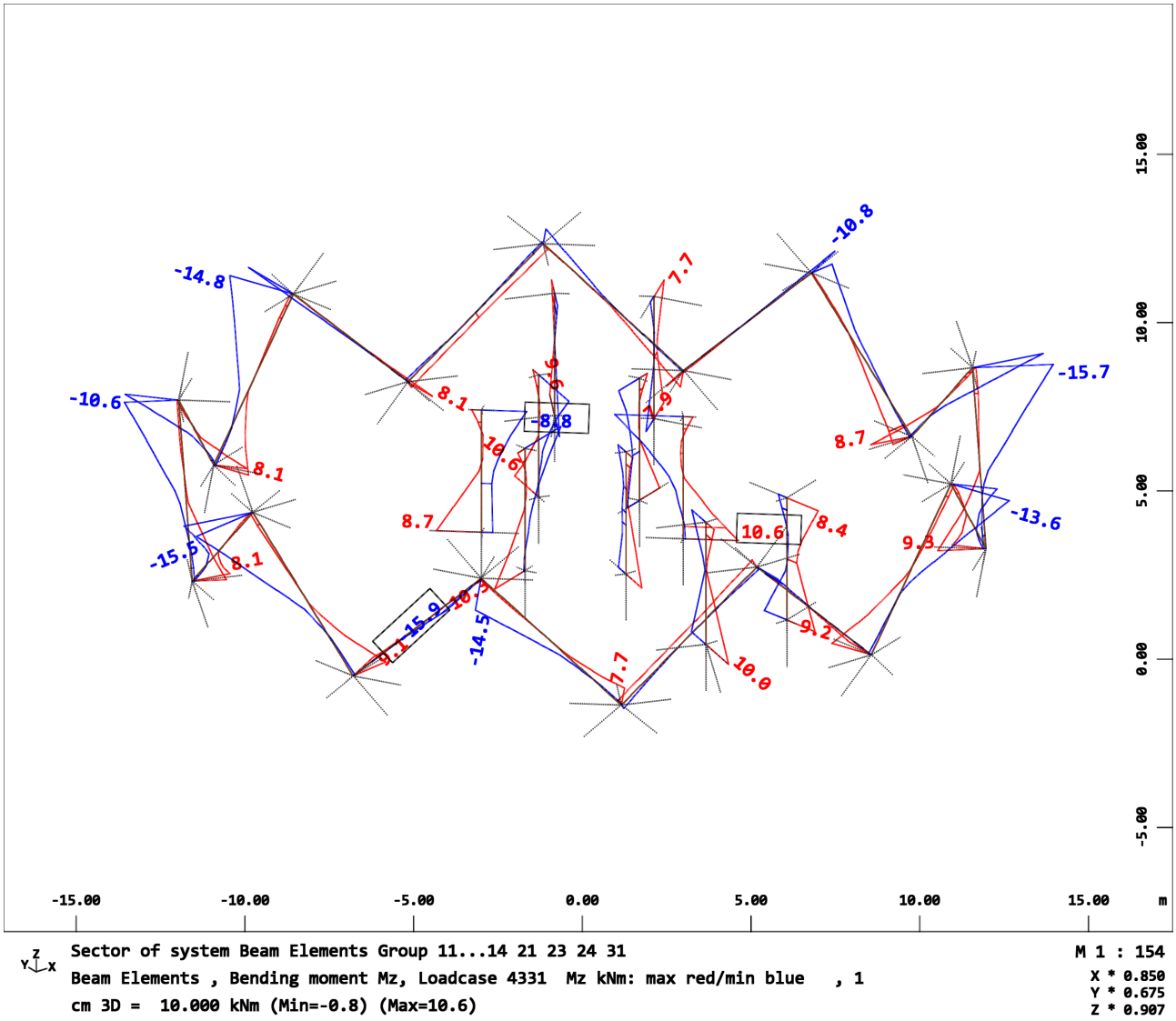
P15_V_L1-L2_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 1 to Level 2 (groups 11, 12, 13, 14, 21, 23, 24, 31) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 158 kN. Envelope extremes: max 29.6, min -263.75 kN.



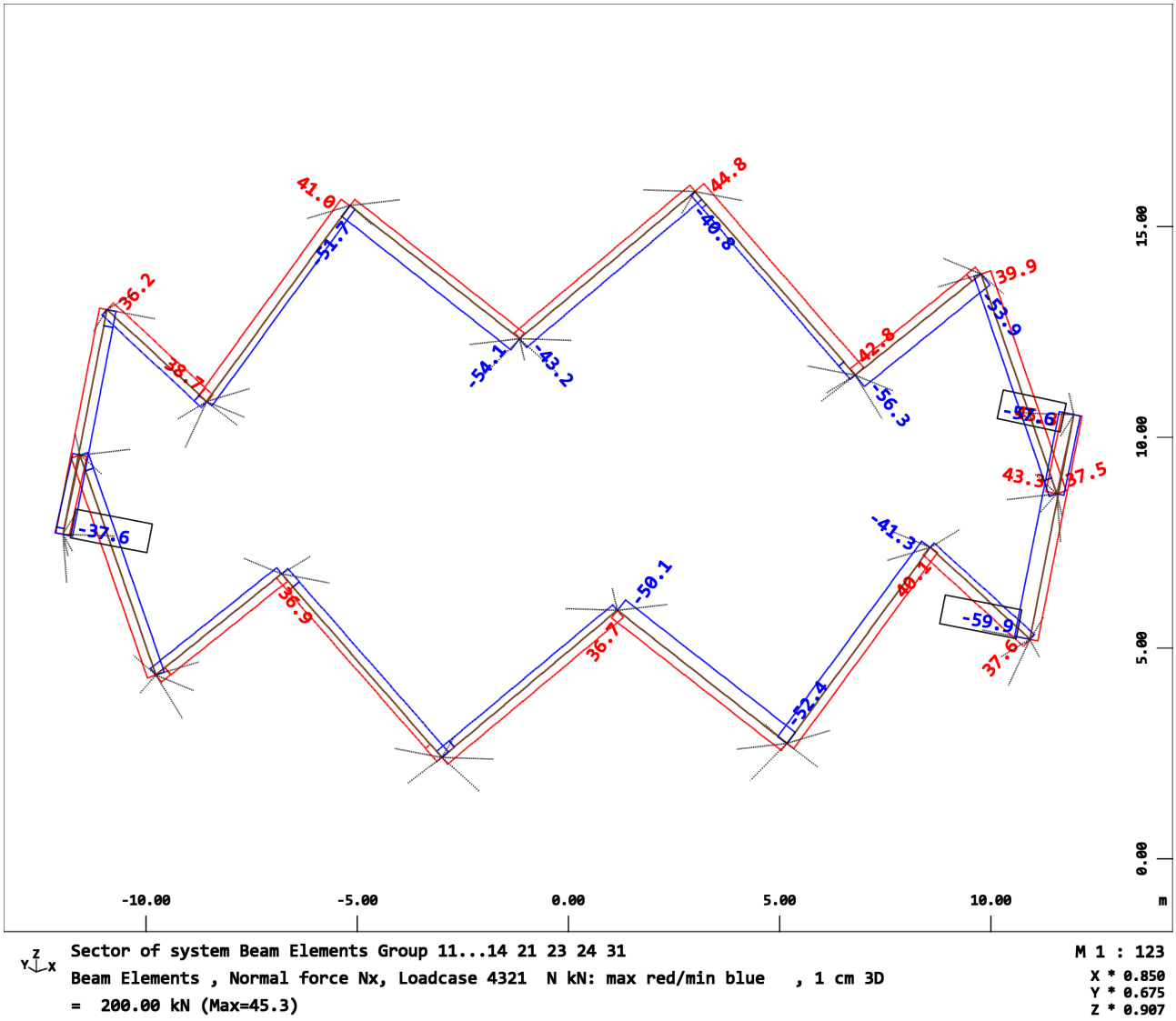
P16_V_L1-L2_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 1 to Level 2 (groups 11, 12, 13, 14, 21, 23, 24, 31) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 10 kNm. Envelope extremes: max 20.94, min -22.34 kNm.



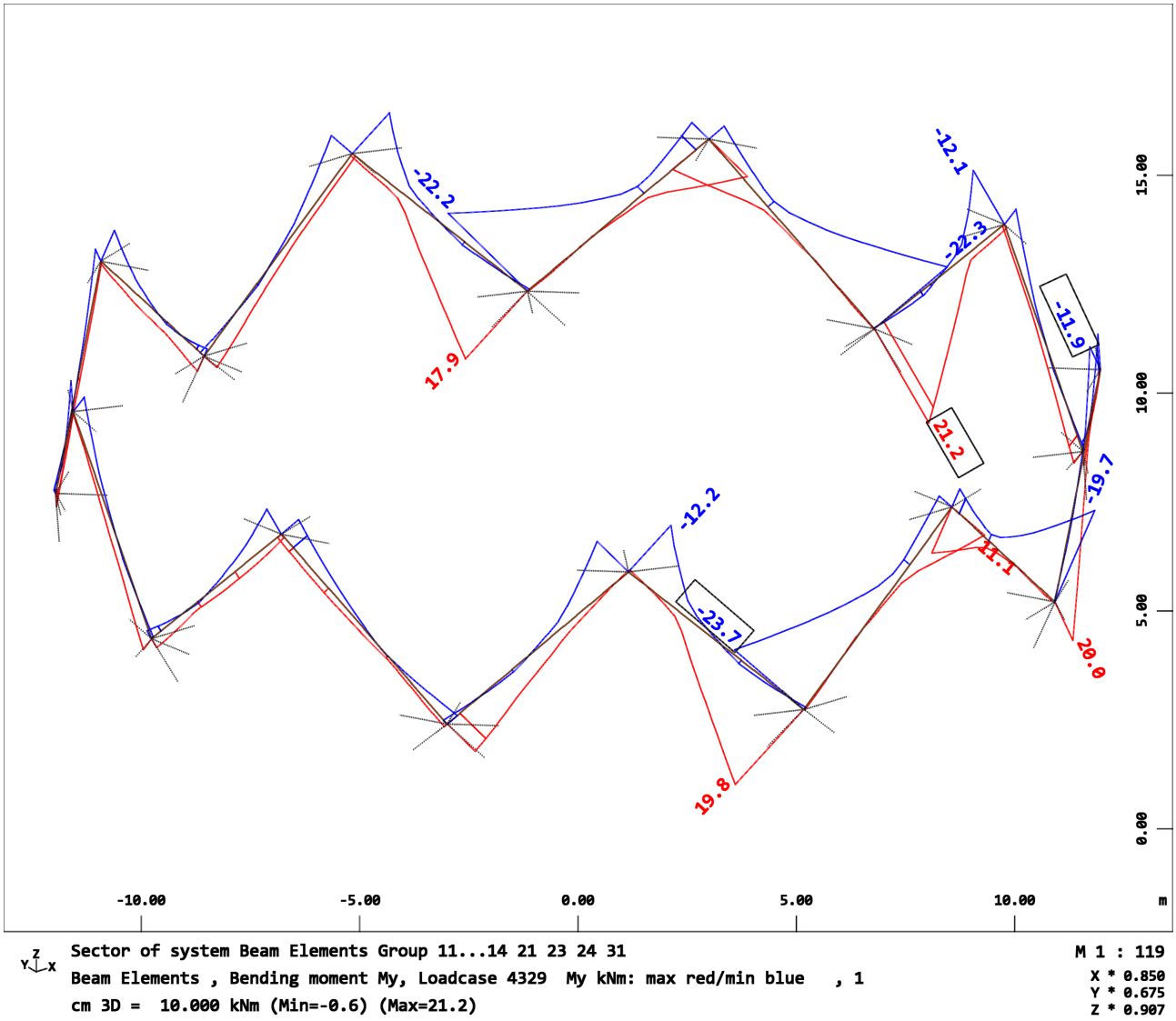
P17_V_L1-L2_Mz

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 1 to Level 2 (groups 11, 12, 13, 14, 21, 23, 24, 31) Mz. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 7.1 kNm. Envelope extremes: max 10.64, min -15.86 kNm.



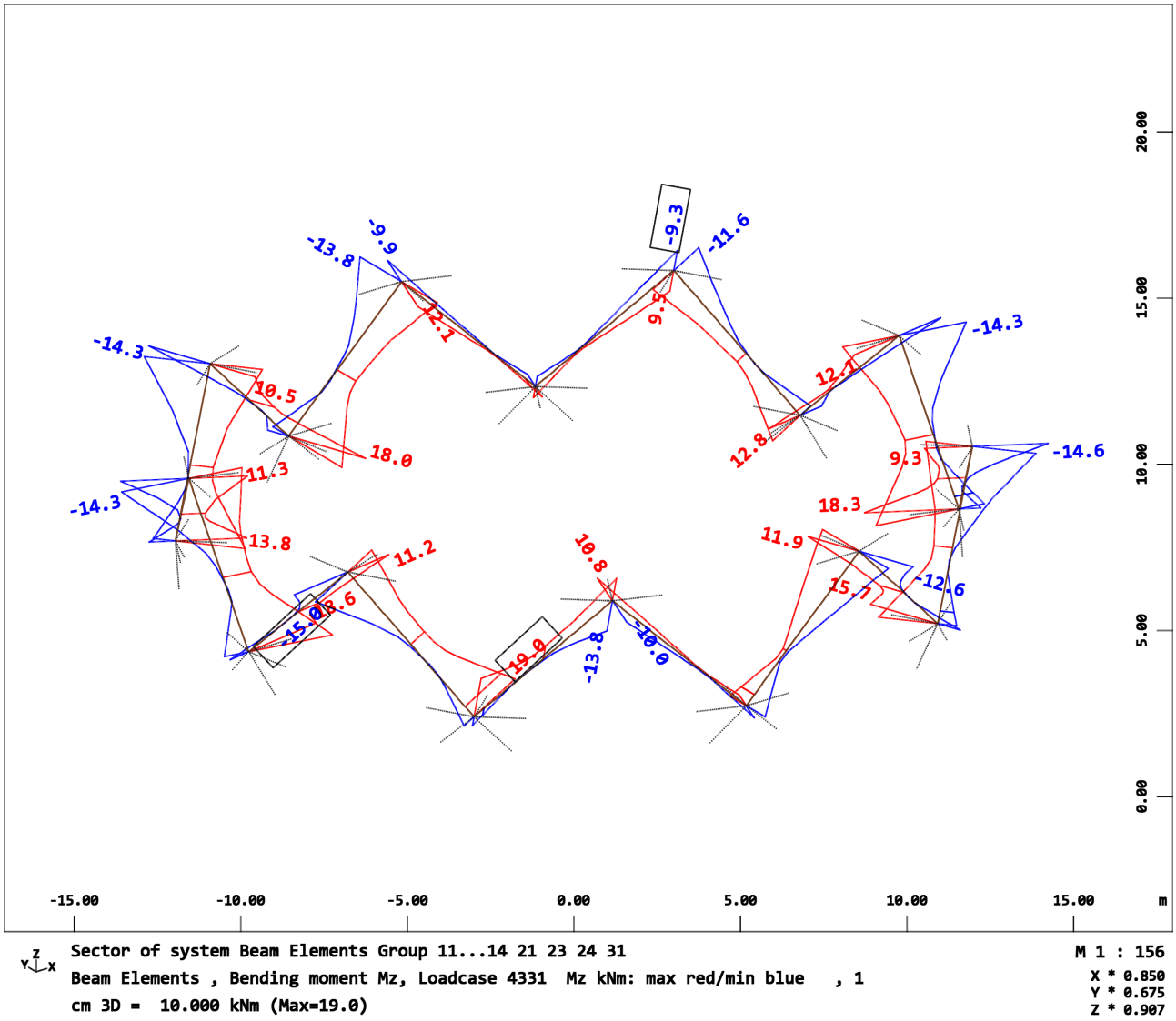
P18_V_L2-RF_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 2 to roof (groups 11, 12, 13, 14, 21, 23, 24, 31) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 36 kN. Envelope extremes: max 45.29, min -59.9 kN.



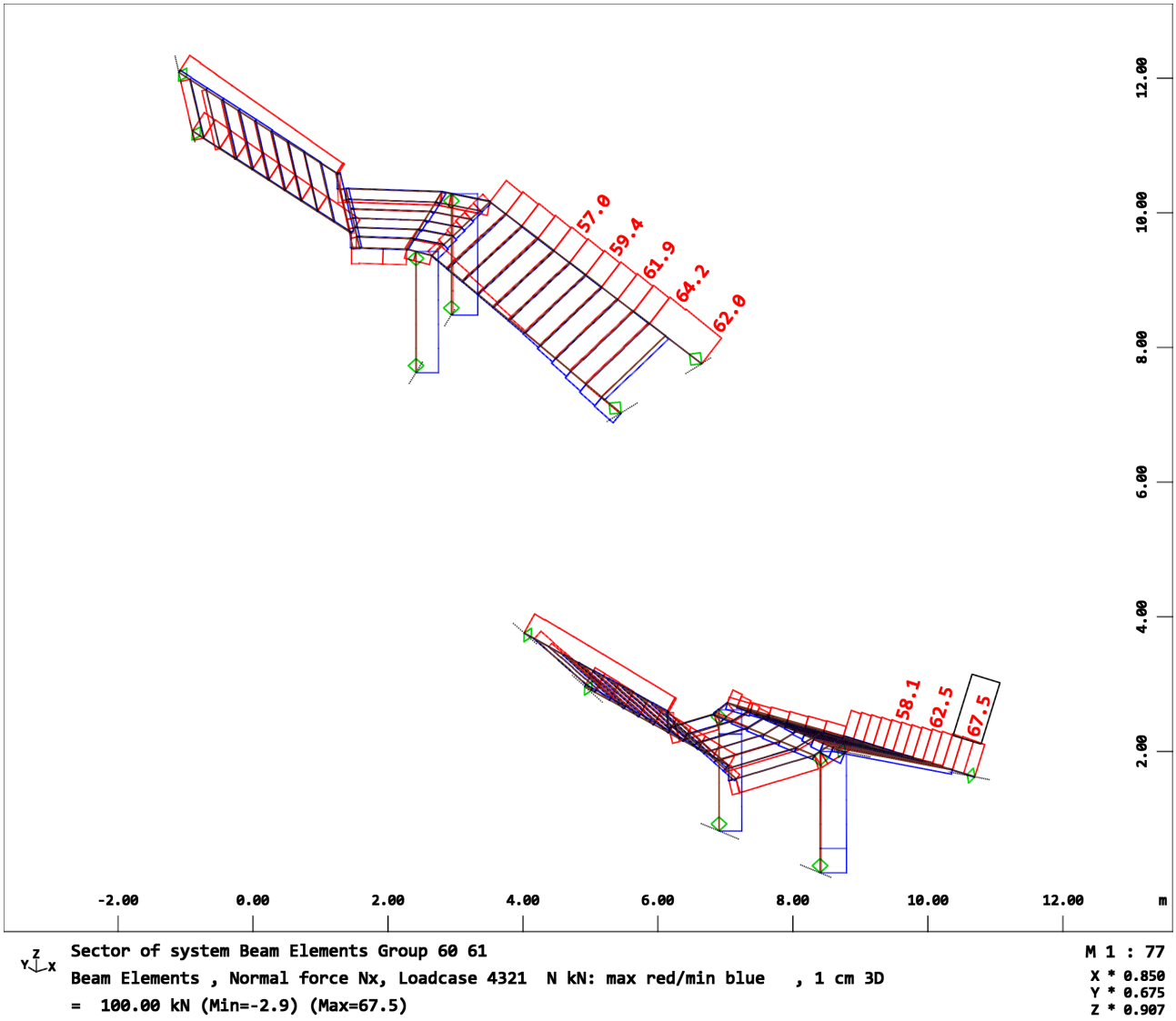
P19_V_L2-RF_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 2 to roof (groups 11, 12, 13, 14, 21, 23, 24, 31) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 11 kNm. Envelope extremes: max 21.15, min -23.68 kNm.



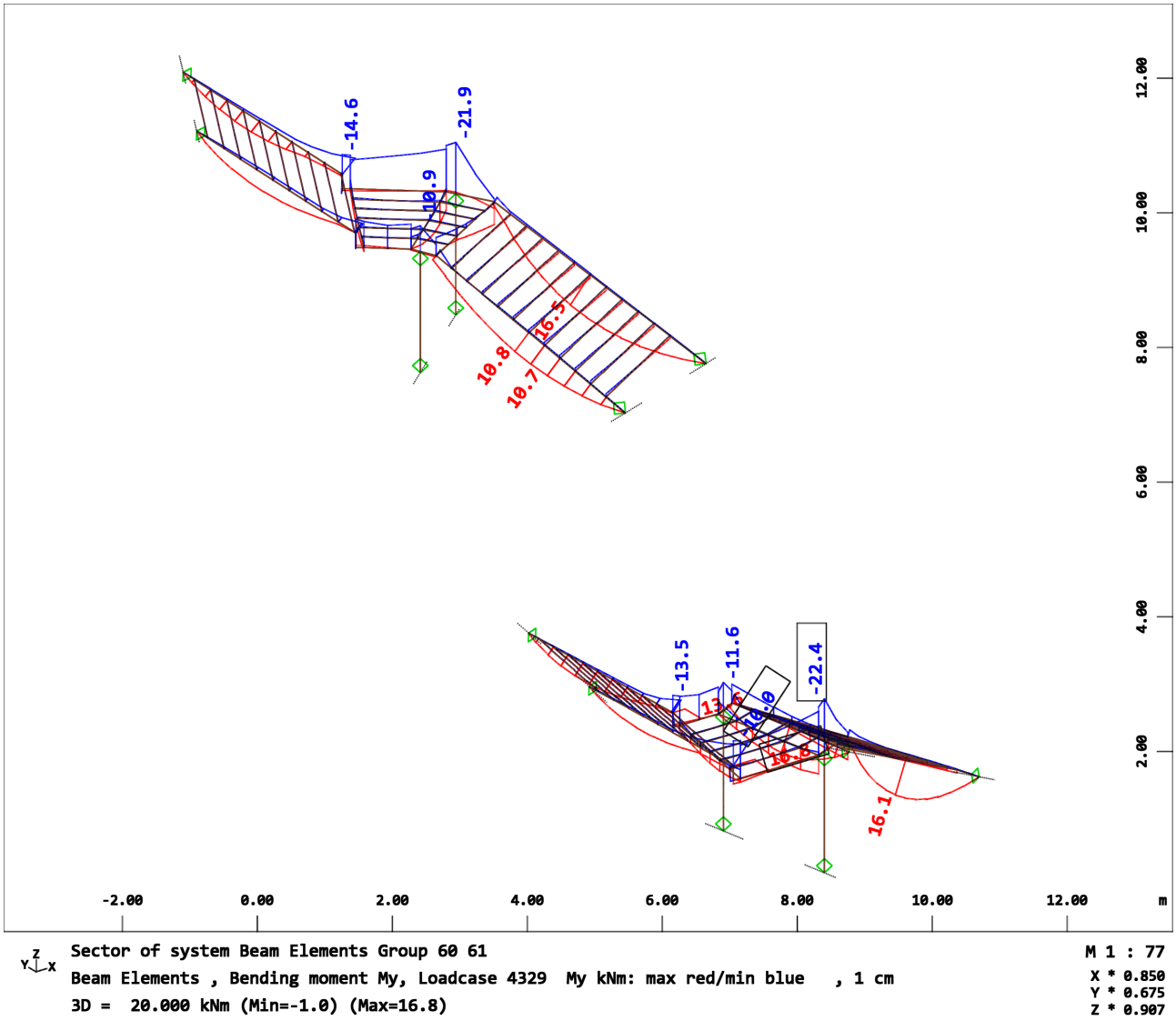
P20_V_L2-RF_Mz

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 2 to roof (groups 11, 12, 13, 14, 21, 23, 24, 31) Mz. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 8.6 kNm. Envelope extremes: max 19.03, min -14.96 kNm.



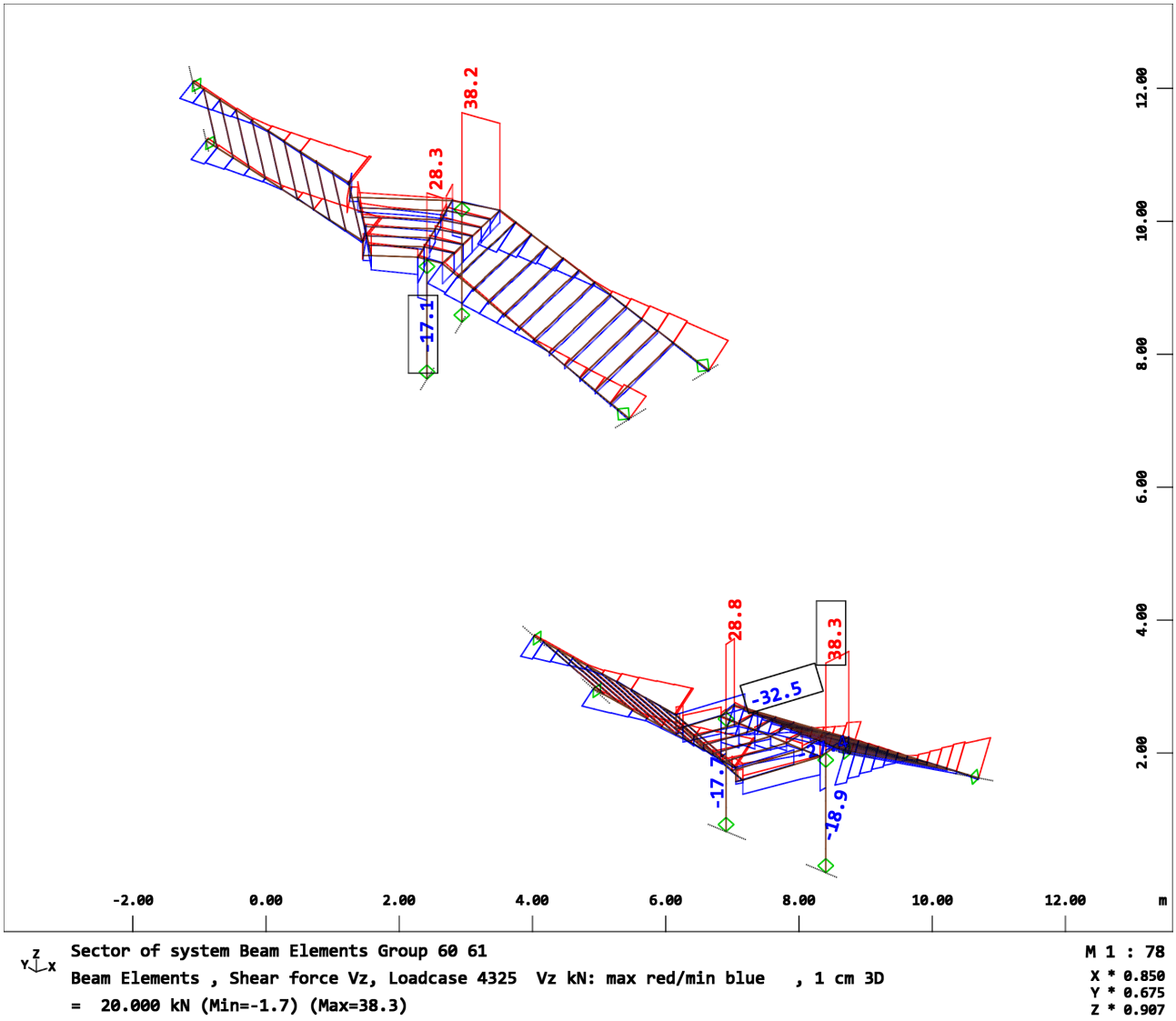
P21_S_int_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. internal stairs (groups 60, 61) N. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 57 kN. Envelope extremes: max 67.49, min -50.94 kN.



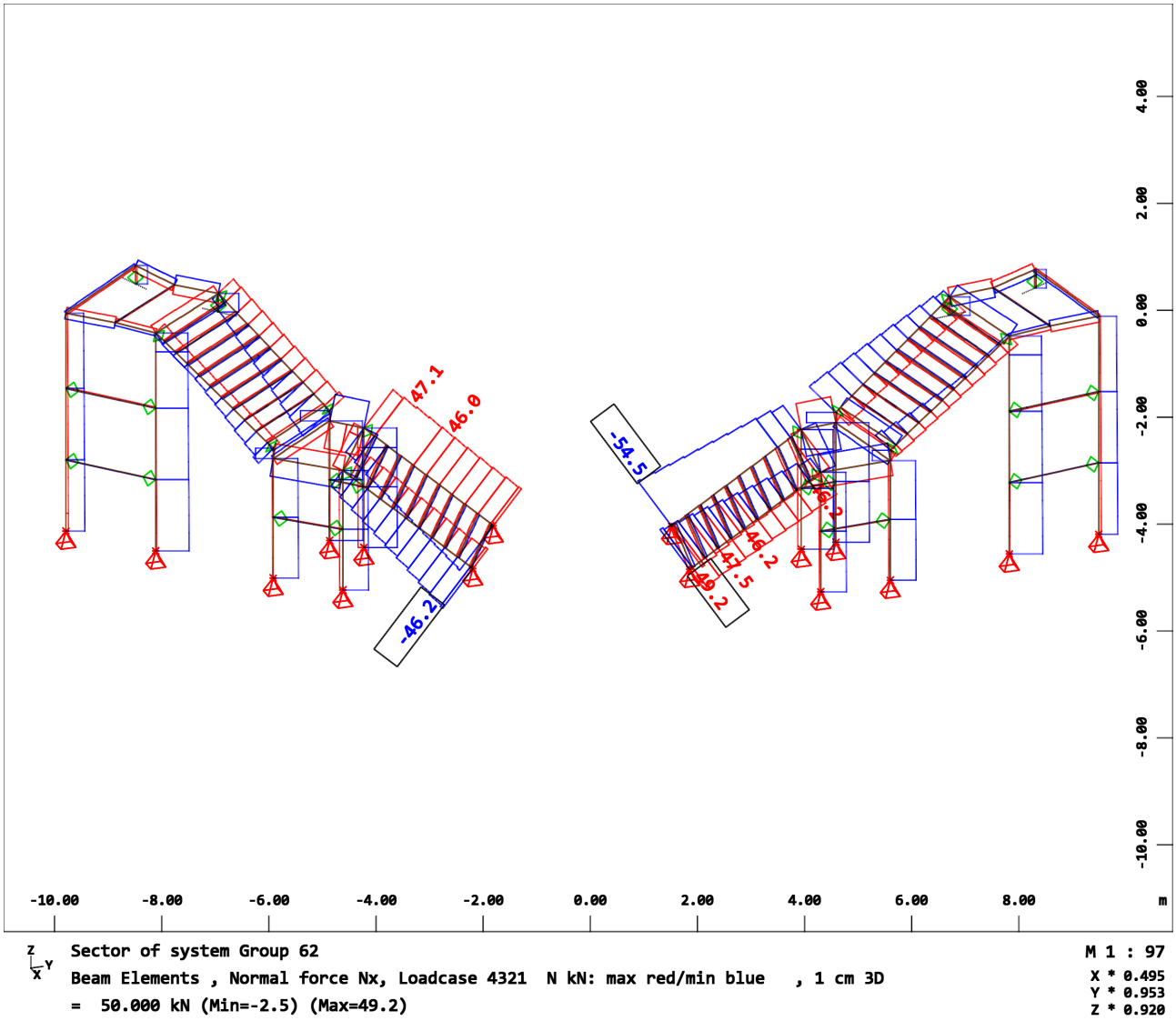
P22_S_int_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. internal stairs (groups 60, 61) My. Diagram scale 1 cm = 20 kNm (on the 25.5 cm sheet); labels only where |value| > 10 kNm. Envelope extremes: max 16.8, min -22.38 kNm.



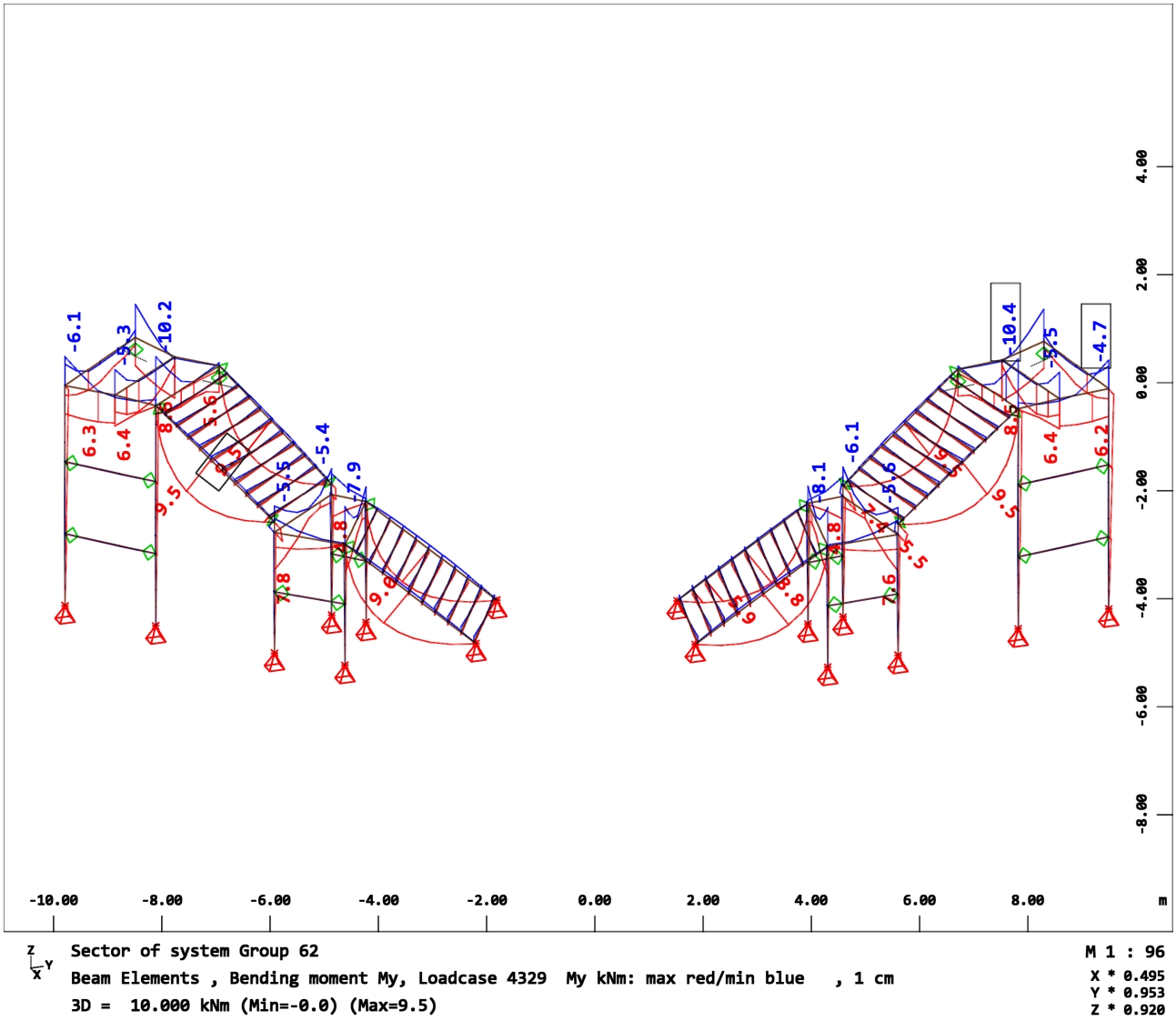
P23_S_int_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. internal stairs (groups 60, 61) Vz. Diagram scale 1 cm = 20 kN (on the 25.5 cm sheet); labels only where |value| > 17 kN. Envelope extremes: max 38.25, min -32.46 kN.



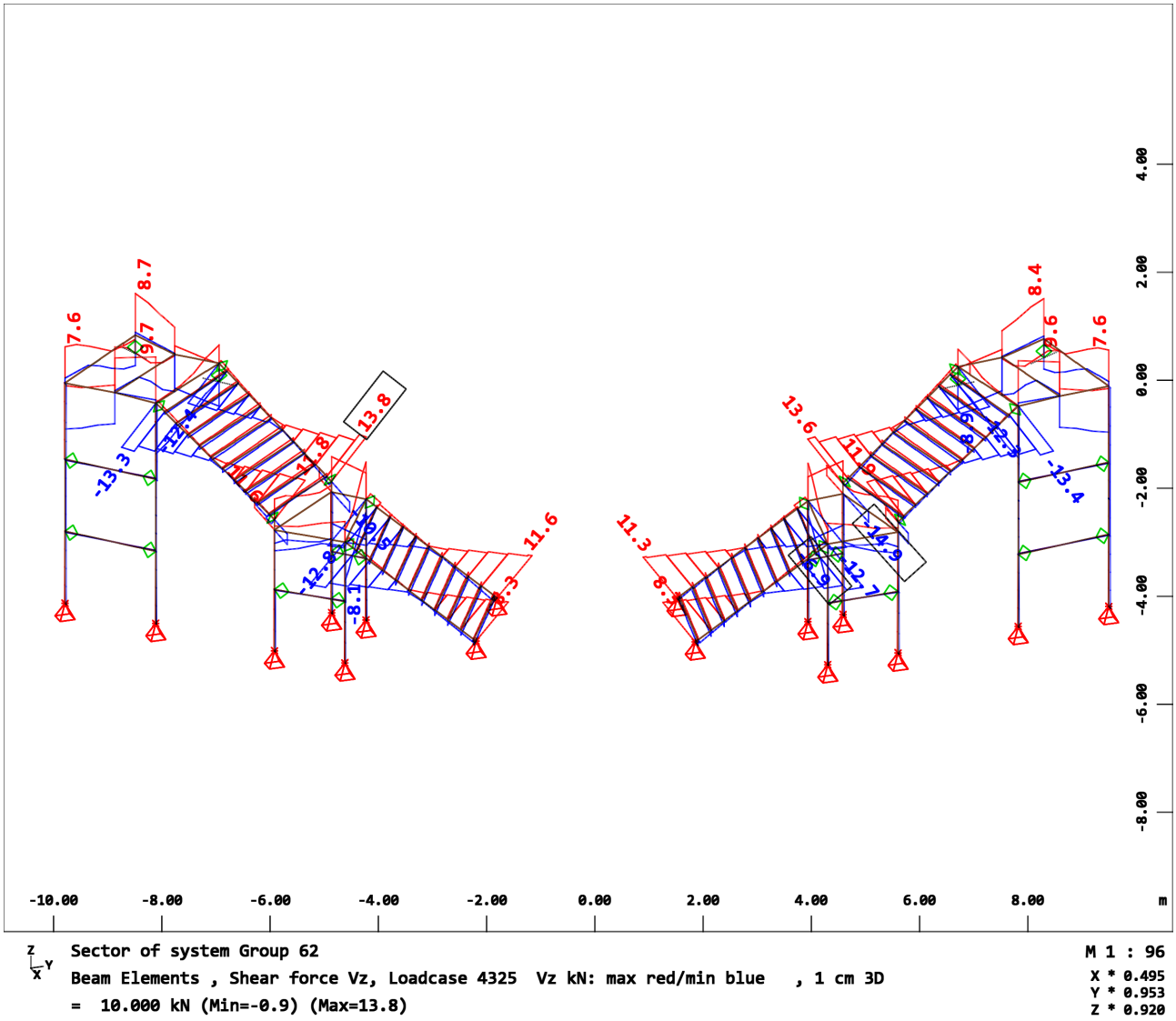
P24_S_ext_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. external stair (group 62) N. Diagram scale 1 cm = 50 kN (on the 25.5 cm sheet); labels only where |value| > 46 kN. Envelope extremes: max 49.23, min -54.52 kN.



P25_S_ext_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. external stair (group 62) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 4.7 kNm. Envelope extremes: max 9.55, min -10.41 kNm.



P26_S_ext_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. external stair (group 62) Vz. Diagram scale 1 cm = 10 kN (on the 25.5 cm sheet); labels only where |value| > 6.7 kN. Envelope extremes: max 13.83, min -14.88 kN.